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Right Hospital, Right Bed, Right Time: A Verified Network Model for Mapping Hospitals and Routing Patients in Albay by Clinical Need, Live Bed Availability, and Ability to Pay

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ABSTRACT

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A patient in Albay usually goes to the nearest hospital, which may have no free bed, may lack the needed service, or may cost more than the family can pay. Patients without an ambulance ride jeepneys, tricycles, and vans whose routes are recorded nowhere. This study built a laptop-computable network model from open data that maps hospitals by capability, services, emergency room, cost class, and reported free beds, and routes each patient, by ambulance or public transport, to the nearest hospital fitting the patient's acuity, required services, payment class, and walking ability. OpenStreetMap roads and estimated headways formed a landmark-anchored timetable graph, e-mail agents gathered bed counts, and an early-stopping search returned the nearest suitable hospital with ranked backups. The model also maps hospital catchments, measures bed access per person and its inequality, assigns many patients to few beds, ranks roads critical to hospital access, and selects sites for new Malasakit Centers. Eighteen results were proven and tested on synthetic instances against independent code. Hospital routing matched the references in all 10,400 queries, and the one-search catchment map agreed at all 11,474 test vertices. Every road closure that delayed patients lay on the predicted shortest-path forest, halving the roads to test. Optimal surge assignment never overfilled a hospital and placed 202 more patients than first-come allocation, while greedy site selection reached at least 82.5 percent of the optimum. The agents rejected all 200 forged replies. Open data therefore suffice to route patients by need, beds, and cost, supporting SDGs 3, 10, and 11. The Provincial Health Office and LGUs are recommended to pilot the model with consenting hospitals and validate it in the field.

Keywords: nearest suitable hospital, time-expanded graph, two-step floating catchment area, minimum-cost assignment, payment gap

1. INTRODUCTION

In 2015 the member states of the United Nations adopted the 2030 Agenda for Sustainable Development, with 17 Sustainable Development Goals (SDGs) and a pledge that no one will be left behind (United Nations General Assembly [UNGA], 2015). Target 3.8 asks for health care that people can use without financial hardship, and target 11.2 asks for transport that is safe, affordable, and accessible for all.

The two targets meet at a single moment: the trip from where a patient is to a hospital that can actually help. In Albay that trip is harder than a map suggests.

The usual answer, “go to the nearest hospital,” ignores all of this. A patient sent to a full hospital loses time at the door.

This study treats the question as a problem in mathematics and computation. Hospitals, roads, transit services, and landmarks become graphs and sets; a patient’s situation becomes a profile; and “the right hospital” becomes the solution of a shortest-path problem over a filtered set of hospitals.

1.1 Statement of the Problem

The study starts from the patient. Table 1 lists the problems that a patient, a family, an ambulance crew, or a health planner in Albay meets on the way to hospital care, together with the part of the model built to deal with each.

Table 1. Problems in reaching hospital care that the study addresses, and the part of the model that answers each.

No.	Problem	What happens to the patient	Part of the model that answers it
P1	The nearest hospital has no free bed	Travels, is turned away, loses time	Live bed status from the agent network; the FULL filter (Defs. 14, 15, 17)
P2	The nearest hospital cannot treat the condition	Referred again after arrival	Service sets of hospitals and the patient’s need set (Defs. 13, 16, 17)
P3	The family cannot pay at the nearest hospital	Delays care or carries debt	Cost classes, payment class, and payment gap (Defs. 13, 17, 20)
P4	An emergency patient is screened by ability to pay	Treatment delayed by a deposit demand	Payment filter switched off for emergencies, as Republic Act No. 10932 requires (Def. 17; Prop. 10)

No.	Problem	What happens to the patient	Part of the model that answers it
P5	No ambulance is available	Must find a jeepney, tricycle, or van route to the hospital and pay the fare	Patient mode on the timetable graph, with the regulated fare (Defs. 4 to 6, 12, 18)
P6	The patient walks slowly because of age, illness, pregnancy, or disability	The usual walking assumptions overstate access	Mobility classes in the patient profile (Def. 16; Prop. 13)
P7	A road is too narrow or a bridge too weak for the ambulance	Crew discovers the obstacle on the way	Ambulance vehicle class on the road graph (Def. 11; Prop. 3)
P8	The first-choice hospital turns the patient away on arrival	No plan for where to go next	Ranked backup hospitals (Def. 18; Prop. 9)
P9	A critical patient is far from the right hospital but near an emergency room	Life-saving care comes late	Stabilization rule (Def. 19)
P10	Many patients need beds at once (typhoon, Mayon Volcano eruption, road crash)	Ambulances converge on the same few beds	Surge assignment as a minimum-cost assignment (Def. 24; Prop. 16)
P11	Flooding or lahar closes roads	Hospitals are cut off without warning	Road segments ranked by how much their loss delays patients (Def. 25; Prop. 17)
P12	Planners do not know which communities are far from affordable beds	Investment goes where need is not greatest	Catchment map, bed-to-population ratio, bed deserts, and site selection (Defs. 21 to 23, 26; Props. 14, 15, 18)
P13	The health office has no live picture of beds	Cannot direct patients during surges	Agent network and bed table with report age (Defs. 14, 15; Prop. 12)

1.2 Objectives of the Study

General objective. The study aimed to build, prove correct, and computationally verify a network model that maps the hospitals of Albay by what they can treat, whether they have a free bed, and what they cost the patient, and that sends each patient, by ambulance or by public transport, to the nearest hospital that meets that patient's actual needs.

Specific objectives. In particular, the study aimed to:

1. build from open data a hospital map of Albay linked to a road graph with vehicle classes, an ambulance class among them, and to a landmark-anchored timetable graph of informal public transport that carries the regulated fares;

2. formulate patient needs as a profile of acuity, required services, payment class, and mobility, and define a suitability rule that decides which hospitals can serve that patient at a given time;
3. design an early-stopping search that returns the nearest suitable hospital and a ranked list of backups in both travel modes, with a stabilization rule for the most critical patients, and prove it exact;
4. build an e-mail agent network that collects bed counts from hospitals, with proven limits on the number of messages and on the age of reports;
5. map hospital catchments and measure bed access and its inequality across anchors and municipalities, including the payment, need, and mobility gaps and a bed-to-population ratio;
6. place many patients in few beds during a surge without overfilling any hospital, as a minimum-cost assignment problem;
7. identify the road segments most critical to hospital access and select sites for new Malasakit Centers or added beds, with proven guarantees for both methods; and
8. verify every proven property numerically against independently written reference code on synthetic instances, and work the full pipeline through a small test network.

1.3 Research Questions

Primary question. Suppose the hospitals and the informal public transport of Albay are described by a machine-readable model built only from public sources. How quickly can a patient starting from any anchor landmark reach a hospital that can treat the patient, has a free bed, and accepts what the patient can pay, and how evenly is that access shared among anchors and among municipalities?

The subsidiary questions follow the specific objectives (Table 2).

Q1. With no official schedules, can departure times be estimated from service headways, and can a time-expanded timetable graph answer earliest-arrival queries exactly and fast enough for every landmark?

Q2. How do limits on width, clearance, gross mass, and axle load change the roads an ambulance or another vehicle may use, and can a larger vehicle ever reach more?

Q3. Under the regulated fares, can routes for a commuting patient be found that balance arrival time, fare, and number of transfers?

Q4. Can a patient's acuity, required services, payment class, and mobility be written as a suitability rule whose behavior is provably monotone, and does the rule obey the Anti-Hospital Deposit Law in emergencies?

Q5. Does an early-stopping search return the exact nearest suitable hospital and its backups in both travel modes, and how much work does it save over a full search?

Q6. Can lightweight software agents keep bed reports fresh through standard e-mail with few messages, honor opt-outs, and discard forged or implausible replies?

Q7. How do bed access and its inequality change when authorized beds are replaced by free beds, by free beds that a poor patient may use, and by free beds per person? Can the Theil index still split inequality into within-municipality and between-municipality parts when the index set is a list of landmarks rather than a full partition of the province?

Q8. When many patients need beds at the same time, can they be placed without overfilling any hospital, and how does that placement compare with letting each patient go to his or her own nearest hospital?

Q9. Which road segments matter most for reaching a suitable hospital, and where would a new Malasakit Center or added beds reach the most people who now lack affordable care?

Table 2. Specific objectives, research questions, and where each is answered.

Specific objective	Research questions	Definitions and results	Verification
1. Hospital map and transport graphs	Q1, Q2, Q3	Defs. 1 to 13; Props. 1 to 3, 7; Cor. 4, 8	Tables 14, 17, 24
2. Patient profile and suitability	Q4	Defs. 16, 17; Prop. 10; Cor. 11	Tables 15, 16, 26, 28

Specific objective	Research questions	Definitions and results	Verification
3. Nearest suitable hospital and backups	Q5	Defs. 18, 19; Props. 9, 13	Tables 15, 16, 18, 26
4. Agent network for bed counts	Q6	Defs. 14, 15; Prop. 12	Table 15
5. Catchments, bed access, and inequality	Q7	Defs. 7 to 10, 20 to 23; Props. 5, 6, 14, 15	Tables 16, 19, 22, 27, 29
6. Surge assignment	Q8	Def. 24; Prop. 16	Tables 16, 30
7. Critical roads and site selection	Q9	Defs. 25, 26; Props. 17, 18	Table 16
8. Verification and worked instance	All	Sections 3.3 to 3.5	Tables 14 to 30

1.4 Review of Related Literature

1.4.1 Measuring Accessibility

In transport research, accessibility means how easily people can reach opportunities such as jobs, schools, markets, and health services. Hansen (1959) offered an early formal measure in which each opportunity counts for less the longer or costlier it is to reach.

The cumulative opportunity measure of Wachs and Kumagai (1973) belongs to the location family. It simply counts the opportunities inside a fixed travel-time budget.

1.4.2 Measuring Spatial Access to Health Care

Counting reachable beds has one blind spot. A hospital with 40 beds counts as 40 for every community that can reach it, whether five hundred people or fifty thousand people depend on those beds.

1.4.3 Equity in the Distribution of Access

Provincial averages can hide sharp inequalities, so that some communities remain underserved even when the overall figure looks acceptable. Lucas et al. (2016) proposed a formal method for judging whether access is fairly shared, and Pereira et al. (2017) linked the distribution of transport benefits to theories of distributive justice.

1.4.4 Hospital Levels, Triage, and What a Patient Needs

Three established practices decide which hospital can treat a given patient. Triage comes first.

The third practice is less formal but matters just as much to the patient: what the patient specifically needs. A woman in uncomplicated labor needs a maternity ward; a newborn in distress needs a neonatal ICU; a patient with kidney failure needs dialysis.

1.4.5 Bed Reporting in the Philippines

During the COVID-19 pandemic the DOH ran DataCollect, through which 1,906 public and private hospitals and infirmaries reported how many COVID-19 beds and mechanical ventilators they had in use. Reporting was 80 to 95 percent complete by April 2020 and 98 percent complete by December 2022 (Cabaro et al., 2023).

1.4.6 Payment for Hospital Care in the Philippines

Three sets of laws shape what a patient pays and where. Under the Universal Health Care Act, every Filipino is automatically enrolled in the National Health Insurance Program, and services in basic or ward accommodation carry no co-payment (Republic Act No. 11223, 2019).

The first two laws tie the choice of hospital to cost. The third removes cost from the decision in an emergency.

1.4.7 Emergency Medical Services and Facility Location

Operations research on emergency medical services has centered on the ambulance, chiefly on where units should be based and how to reposition them as calls arrive (Brotcorne et al., 2003). The classic tool is the maximal covering location problem of Church and ReVelle (1974): choose a fixed number of sites so that as much demand as possible lies within a service-time standard of at least one site.

The receiving end of the trip, the hospital, gets less attention in this literature. It is the focus here.

1.4.8 Network Flows and Assigning Patients to Beds

When several patients need beds at once, the beds become a shared, limited resource. Sending every patient to his or her own nearest hospital can overload one facility while beds stay empty at the next.

1.4.9 Road Vulnerability and Catchments on Networks

Jenelius et al. (2006) measured how important a road link is by how much travel cost rises when the link is closed, and how exposed a place is by how much its own travel cost rises. For hospital access in Albay the question is concrete.

1.4.10 Journey Planning on Timetable Networks

The time-expanded model assigns a node to each departure and each arrival and turns the search for the best journey into a shortest-path problem on a fixed directed graph. The time-dependent model, in contrast, uses one node per stop, and the arc travel times depend on the clock (Pyrga et al., 2008).

1.4.11 Open Data on Informal Transport

Most transit data in circulation follow the General Transit Feed Specification (GTFS). TriMet, the transit agency of Portland, Oregon, developed the format with Google in 2006, and its upkeep has since passed to MobilityData, a nonprofit (MobilityData, n.d.-a).

A school research team can take two lessons from this record. The first is widely noted: the feed tends to be the longest-lived output.

1.4.12 Landmarks as Units of Analysis

Grouping results by administrative zone brings in Openshaw’s (1984) modifiable areal unit problem: the answer changes with the way the zones are drawn. Albay’s 720 barangays (Philippine Statistics Authority [PSA], n.d.) differ enormously in size.

The measure used in this paper avoids zones. Its index set is a finite list of real places, each named in at least one open register: a terminal, a public market, a municipal hall, a district hospital, a public high school.

1.4.13 The Philippine and Albay Setting

Route planning in the Philippines is now partly a local responsibility. The Omnibus Franchising Guidelines of the Department of Transportation (DOTr, 2017), the basis of the Public Utility Vehicle Modernization Program, require each local government unit (LGU) to draw up a Local Public Transport Route Plan with evidence behind each proposed route, and the Department of

the Interior and Local Government spelled out how LGUs should prepare such a plan (DILG, 2018).

The province has 18 LGUs with 720 barangays in all (PSA, n.d.), close to 2,575 km² of land (PhilAtlas, n.d.), and 1,374,768 residents at the 2020 census (PSA, 2021). Mayon Volcano and the yearly typhoon season make road closures and sudden surges in patients a regular part of life there.

1.4.14 Software Agents over E-mail

Wooldridge and Jennings (1995) define an agent as software that acts on its own in an environment, notices and responds to changes there, takes initiative toward its goals, and communicates with other agents. When a task is split among several agents, one part may fail without bringing down the others.

1.5 Research Gap

The review leaves six gaps. Four concern the patient and the hospital; two concern the transport that links them.

- **(G1) No live, needs-aware hospital map.** Official registers record a hospital's level and its authorized beds, not its free beds today or its specific services. An ordinary map can locate hospitals but cannot say which of them can treat this patient, has a bed now, and accepts what the family can pay. The hospital ready to admit a patient is often not the one nearest to the patient's family or to the ambulance crew. Closing this gap matters most for SDG target 3.8 and for low-income families.
- **(G2) No treatment of many patients at once.** Nearest-hospital advice is given one patient at a time. During typhoons, eruptions, and road crashes, many patients compete for the same few free beds, and no open model places them without overfilling a hospital.
- **(G3) No measure of bed access that counts crowding, cost, and need.** Provincial and municipal plans describe health access by counts of facilities and beds. No figure tells a planner how many free, affordable, and suitable beds each community can reach in a given time, per person, or how unevenly that is shared.

- **(G4) No guidance on where to add care or which roads to protect.** Without the measures of G3, there is no principled way to choose where a Malasakit Center or extra beds would help most, or which road closures would cut patients off from care.
- **(G5) No dataset of informal transport.** No machine-readable routes, stops, or headways exist for any part of Albay. Claims about access are therefore missing or rest on straight-line distance, which overstates access where terrain, water, or the road hierarchy forces a detour.
- **(G6) No model that knows the vehicle.** The loads that Albay's rural roads can carry vary widely. Some roads are too narrow, too steep, or too rough, or carry a bridge with a low limit. A provincial bus or a six-wheel truck cannot use them, even though tricycles and cars use them every day, and an ambulance crew needs to know which group its van belongs to.

Earlier studies of informal transport relied on manual field mapping (Section 1.4.11). None asked what could be proven and computed when direct field collection is not yet possible, and none carried the result through to the hospital bed.

1.6 Contributions of the Study

- **C1. A hospital map with needs, beds, and cost.** Section 2.7 defines a hospital record that carries capability level, service set, emergency room, cost class, and contact address, all drawn from public registers, and Section 2.8 adds a live bed status gathered by software agents.
- **C2. A patient profile and a suitability rule.** Section 2.9 writes a patient's situation as acuity, required services, payment class, and mobility. Proposition 10 shows that the rule never offers more hospitals to a patient who needs more, and that it never filters an emergency patient by ability to pay.
- **C3. An exact hospital search with backups.** Proposition 9 proves that an early-stopping search returns the exact nearest suitable hospital and a ranked list of backups, and Proposition 13 proves that slower walking never shortens a patient's trip, so the mobility gap is never negative.

- **C4. Catchments and bed access per person.** Proposition 14 builds the whole hospital catchment map in a single search, and Proposition 15 shows that the bed-to-population ratio never counts a bed twice.
- **C5. Surge assignment.** Proposition 16 shows that many patients can be placed in few beds optimally, in whole numbers, and without overfilling any hospital.
- **C6. Planning tools with guarantees.** Proposition 17 proves that only road segments on the shortest-path forest can delay patients when closed, and Proposition 18 guarantees that greedy site selection reaches at least 63.2 percent of the best possible coverage.
- **C7. A verified transport layer.** Sections 2.1 to 2.6 build the landmark-anchored timetable graph, the vehicle classes, and the fare model, and Propositions 1 to 8 prove the query engine correct and the inequality and fare properties sound.
- **C8. A repeatable, open pipeline.** Every input links back to a named public register (Appendix A), every result is reproduced by a script with a fixed random seed, and the synthetic feed can later be replaced by a field-verified feed without changing the mathematics. Creating the feed and the hospital map supports the open-data aims of SDG target 17.18.

Of the eighteen numbered results in Section 3.2, Propositions 1, 2, 5, and 6 restate established results for a landmark setting, with their sources named in their headings, and Proposition 18 applies the theorem of Nemhauser et al. (1978) to a coverage function defined here. The rest, namely Propositions 3, 7, 9, 10, and 12 to 17 and Corollaries 4, 8, and 11, are the researchers' own.

1.7 Scope and Delimitation

The study covers the province of Albay: its 18 LGUs, three legislative districts, and the hospitals and infirmaries listed in public registers. It is entirely computational.

The model routes patients; it does not diagnose them. The acuity level and the required services are entered by an ambulance crew or by the user, and the attending physician's judgment always overrides the application.

1.8 Relation to the Sustainable Development Goals

The goals are interconnected, and one piece of work can bear on several of them (UNGA, 2015). This is a computational study and a planning tool; it has not been applied in practice, and no indicator value for Albay is reported here.

The closest goal is SDG 3, Good Health and Well-being. Hospital mode aims to bring each patient to the nearest hospital that offers the required level and service and has a free bed, without letting cost decide an emergency.

SDG 11, Sustainable Cities and Communities, enters through target 11.2, which asks for safe, affordable, and accessible transport for all, with special attention to the needs of those in vulnerable situations. The landmark measure relates to indicator 11.2.1 (UNGA, 2017) but differs from it: the indicator counts who has convenient access to public transport, while the landmark measure counts what that transport reaches within 30 minutes.

SDG 10, Reduced Inequalities, appears in the measures of Section 2.4. Targets 10.2 and 10.3 concern inclusion and less unequal outcomes (UNGA, 2015).

Some features may also serve other goals if the tool is validated and adopted: vehicle-class routing and critical-road analysis for infrastructure and disaster response (target 9.1 and target 11.5), the bed table during typhoons and Mayon Volcano alerts (target 13.1), the payment-gap map and site selection for services to the poor (target 1.4), the school layer (target 4.a), and the open dataset (target 17.18). These are plausible applications, not proven outcomes.

2. FRAMEWORK

The framework has two layers. The transport layer (Sections 2.1 to 2.6) describes how a person moves through Albay by road, on foot, and on informal public transport, and what that costs.

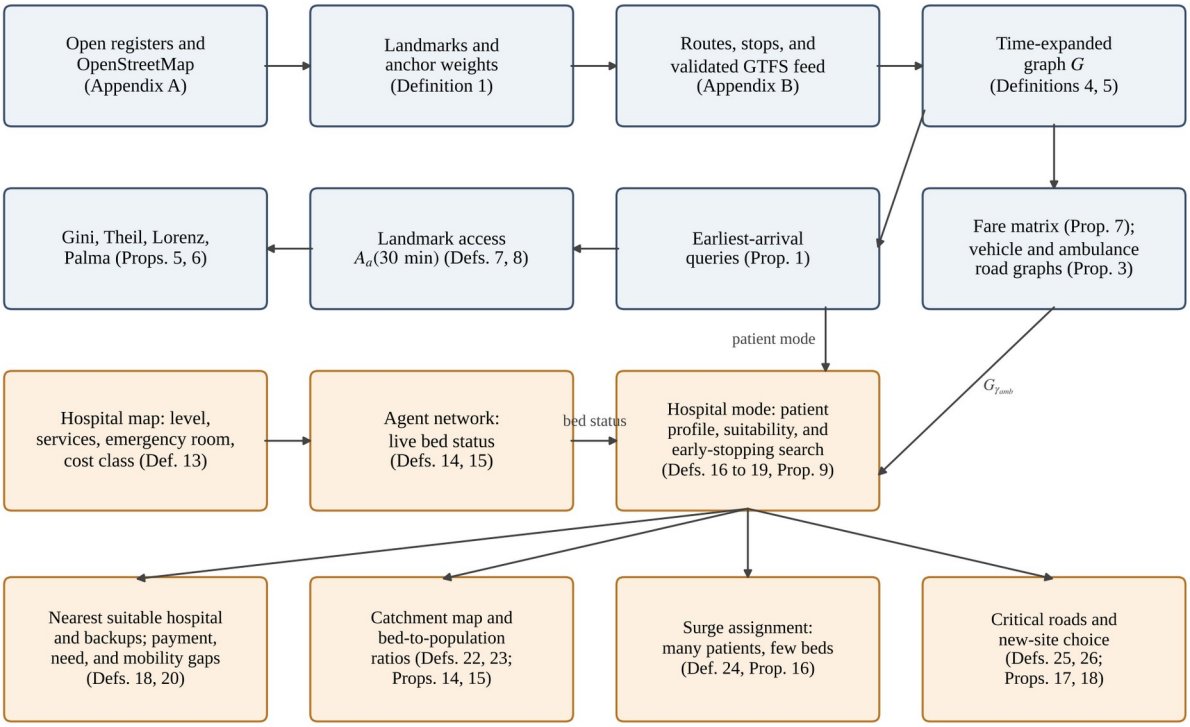


Figure 1. The model from open data to patient recommendations. Blue boxes form the transport layer; orange boxes form the hospital layer, which depends on the agent network for bed status.

2.1 Study Area, Landmarks, and Spatial Units

The study area is the province of Albay, all 18 of its LGUs in its three legislative districts (Table 3). The First District covers the eastern seaboard municipalities, the Second District the capital, Legazpi City, and the municipalities around it, and the Third District the more inland municipalities.

Table 3. The 18 LGUs of Albay, with 2020 population and land area. Population: PSA (2021).

District	**Local Government Units**	**Population (2020)**	**Land Area (km²)**
First	Tabaco City, Bacacay, Malilipot, Malinao, Santo Domingo, Tiwi	395,907	547.88
Second	Camalig, Daraga, Legazpi City, Manito, Rapu-Rapu	477,781	673.85
Third	Guinobatan, Jovellar, Libon, Ligao City, Oas, Pio Duran, Polangui	501,080	1,353.18
Total	18 units	1,374,768	2,574.91

Definition 1 (Landmark set). A landmark is a fixed place with a known name and general public importance, whose location can be pinned down and which appears in at least one public register. The landmark set is

$$L = L_a \cup L_d \tag{1}$$

Source: formulated in this study (Definition 1)

where L_a is the set of anchors and L_d the set of essential destinations. Anchors are the places where people without private vehicles plausibly begin a trip: a public transport terminal, the municipal or city hall, the public market, the town plaza, a port or pier, or a major junction where drivers usually pick up passengers.

Each anchor $a \in L_a$ is represented by o_a, the road-network node nearest to its register coordinate. Its demand weight P_a comes from the barangays assigned to it.

Each destination $p \in L_d$ belongs to one service category $k \in K$ and has a size w_p taken from its register: service windows for an office, classrooms for a school, beds for a hospital. $L_d(k)$ denotes the destinations of category k. Hospitals need more attributes than a size, because a bed count alone does not say whether the hospital can serve a given patient now.

Table 4 lists the register used for each landmark category. Appendix A explains how each source is obtained, and Appendix C gives the inclusion rules and the fields recorded for each landmark.

Table 4. Landmark categories, their role in the model, and the open registers used to identify them. Every category can be compiled from existing records without field visits.

Landmark Class	Role	Open Register
Public transport terminal	anchor	OpenStreetMap; route-plan annexes of LGUs
Municipal or city hall	anchor	OpenStreetMap; Philippine Standard Geographic Code
Town plaza, port, or pier	anchor	OpenStreetMap; port list of the Philippine Ports Authority
Major boarding junction	anchor	Highway junction nodes in OpenStreetMap
Hospital, infirmary, rural health unit	destination	National Health Facility Registry (DOH); hospital licensing records; PhilHealth accreditation list; Malasakit Center directory (DOH)
Public elementary and secondary school	destination	Masterlist of Schools, Department of Education
Government service office	destination	OpenStreetMap; agency directories

Landmark Class	Role	Open Register
Public market	anchor and destination	OpenStreetMap; LGU lists
Designated evacuation center	destination	LGU disaster risk reduction plans

Sources: OpenStreetMap contributors (2026) and PSA (n.d.) for the anchors; for destinations, the Department of Education (2021), the DOH (n.d.-a, n.d.-b, 2012), and the Philippine Health Insurance Corporation (2026); ports from the Philippine Ports Authority (n.d.). Appendix A gives access details.

Barangays matter to the model only because their populations make up the anchor weights. No barangay receives an accessibility value of its own, no barangay centroid serves as an origin, and nothing is claimed about any individual barangay.

2.2 Network Definitions

Definition 2 (Road network). $G_R = (V_R, E_R)$ is a weighted directed graph taken from OpenStreetMap (OpenStreetMap contributors, 2026). An arc e has a length $\ell_e > 0$ and a free-flow speed v_e, and its attribute vector $a_e = (a^w_e, a^h_e, a^m_e, a^x_e)$ gives four limits for the arc: usable width, vertical clearance, maximum gross mass, and maximum axle load. An attribute that was never surveyed receives the value $+\infty$ and a flag.

Definition 3 (Walking network). $G_W = (V_R, E_W)$ reuses the vertices of the road network. Walking an arc takes ℓ_e / v_{walk}, where $v_{walk} = 4.5$ km/h is a design value inside the range of normal adult walking speeds in the meta-analysis of Bohannon and Williams Andrews (2011). A single walking leg is capped at $\bar{t}_W = 15$ min. Section 2.9 lets both values depend on the patient.

Definition 4 (Transit network). Boarding points form the set S. One run of a vehicle visits a sequence of stops, and that sequence is a trip. A connection $c = (c_{dep}, c_{arr}, \tau_{dep}(c), \tau_{arr}(c), trip(c))$ records that some vehicle leaves stop c_{dep} at time $\tau_{dep}(c)$ and reaches the next stop c_{arr} at time $\tau_{arr}(c)$. All connections of the service day make up C. Changing to a vehicle on a different trip costs at least the change time θ_{ch}.

Definition 5 (Time-expanded graph, following Pyrga et al., 2008). The time-expanded graph is $G = (V, E)$, in which each vertex is a stop-time pair (s, t) that stands for a departure event or an arrival event. Arcs are of four kinds: ride arcs from the departure event of a connection to its own arrival event; wait arcs between consecutive departure events at the same stop; transfer arcs of duration θ_{ch} from an arrival event to the earliest departure event at the same stop that it can still

catch; and walk arcs from an arrival event at *s* to the earliest departure event at *s′* reachable on foot, with duration equal to the walking time. Every arc satisfies *t*(head) ≥ *t*(tail).

2.3 Travel Time and Accessibility

Definition 6 (Earliest arrival, following Pyrga et al., 2008). Fix an origin *o,* a destination *d*, and a departure time t_0, and let $J(o, d, t_0)$ be the set of feasible journeys from *o* to *d* that leave no earlier than t_0. Then

$$\tau^*(o, d, t_0) = \min\{\, \text{arrival time of } j : j \in J(o, d, t_0) \,\} \tag{2}$$

Source: Pyrga et al. (2008)

with $\tau^* = +\infty$ when $J(o, d, t_0)$ is empty. The door-to-door travel time is $T(o, d, t_0) = \tau^*(o, d, t_0) - t_0$.

Informal operators publish no departure times, so *T* shifts with whichever departure instant happens to be sampled. For that reason, atlas-level results are averages over a grid of departure times.

$$\bar{T}(o, d) = \frac{1}{|\mathcal{T}_0|} \sum_{t_0 \in \mathcal{T}_0} T(o, d, t_0), \qquad \mathcal{T}_0 = \{06{:}00,\ 06{:}05,\ \ldots,\ 08{:}00\} \tag{3}$$

Source: averaging as in Owen and Levinson (2015); see also Stępniak et al. (2019)

Definition 7 (Cumulative landmark accessibility, adapted from Wachs & Kumagai, 1973). For an anchor *a*, a destination category *k* from Table 4, and a travel-time budget θ,

$$A_a^k(\theta) = \sum_{p \in L_d(k)} w_p \cdot \mathbf{1}\left[\, \bar{T}(o_a, p) \le \theta \,\right] \tag{4}$$

Source: adapted from Wachs and Kumagai (1973); the anchor replaces the residential zone

Equation (4) keeps the form of Wachs and Kumagai's (1973) cumulative opportunity measure. Only the origin differs, since an anchor of Definition 1 takes the place of the residential zone.

Definition 8 (Impedance-weighted accessibility, based on W. G. Hansen, 1959). As a check, a second form discounts each destination by a negative exponential of its travel time:

$$A_a^{k,\beta} = \sum_{p \in L_d(k)} w_p \exp\!\left(-\beta \bar{T}(o_a, p)\right), \qquad \beta > 0 \tag{5}$$

Source: W. G. Hansen (1959); exponential form as reviewed by Handy and Niemeier (1997)

Equation (5) has no threshold, so it accompanies the cumulative measure wherever the sharp step at θ affects a result (Section 3.5).

2.4 Measures of Access Inequality

Consider one destination category. Let $A = (A_1, \ldots, A_{Na})$ be the vector of anchor scores, with anchor weights P.

Definition 9 (Population-weighted Gini coefficient, from Gini, 1912). Cowell (2011) gives the weighted form below, the same form that Pereira et al. (2017) and Lucas et al. (2016) use:

$$G(A, P) = \frac{\sum_i \sum_j P_i P_j |A_i - A_j|}{2 W^2 \mu} \tag{6}$$

Source: Gini (1912), in the weighted form of Cowell (2011)

Definition 10 (Theil index, from Theil, 1967). With $p_a = P_a / W$,

$$T(A, P) = \sum_a p_a \frac{A_a}{\mu} \ln \frac{A_a}{\mu}, \qquad A_a \geq 0, \quad 0 \ln 0 = 0 \tag{7}$$

Source: Theil (1967)

For the Lorenz (1905) curve of access, anchors are sorted by A_a, and the running share of weight is plotted against the running share of reachable destinations. The Palma ratio (Cobham & Sumner, 2013; Palma, 2011) divides the access of the best-served tenth of the weighted population by that of the worst-served four-tenths; people without statistical training generally find it easier to read than a Gini value (Cobham & Sumner, 2013).

2.5 Vehicle Classes and the Ambulance

Definition 11 (Vehicle class and feasible subgraph). A vehicle class is a vector $\gamma = (\gamma^w, \gamma^h, \gamma^m, \gamma^x)$ whose components give, in order, the width the vehicle needs, the clearance it needs, its gross mass, and the load on each axle. The vehicle may use the arcs in

$$E_\gamma = \{\, e \in E_R : a_e^w \geq \gamma^w,\ a_e^h \geq \gamma^h,\ a_e^m \geq \gamma^m,\ a_e^x \geq \gamma^x \,\} \tag{8}$$

Source: formulated in this study (Definition 11)

so that $G_\gamma = (V_R, E_\gamma)$ is its feasible subgraph, with shortest-path distance d_γ. Writing $\gamma \leq \gamma'$ means that every component satisfies the inequality.

The order is only partial. Every limit of the sedan is at most the matching limit of the sport utility vehicle, so the two are comparable.

Table 5. Vehicle classes used for routing. The entries are representative design values chosen by the researchers, not legal limits.

Class	Width (m)	Clearance (m)	Gross Mass (t)	Axles	Axle Load (t)
Pedestrian	0.8	2.0	–	–	–
Motorcycle, tricycle	1.3	2.0	0.6	2	0.30
Sedan	1.8	1.6	2.0	2	1.00
Sport utility, pickup	2.0	2.0	3.0	2	1.50
Ambulance (van type)	2.0	2.6	3.5	2	1.75
Jeepney, UV van	2.1	2.6	4.5	2	2.25
Four-wheel truck	2.2	3.2	7.5	2	3.75
Six-wheel truck	2.5	3.8	15.0	3	5.00
Ten-wheel truck	2.5	4.2	26.0	5	5.20
Provincial bus	2.6	4.0	18.0	3	6.00

2.6 Fare Model: What the Commuting Patient Pays

A patient without an ambulance pays a fare on every leg, and for a poor family the fare is part of the cost of care.

Definition 12 (Regulated distance fare; Daguno-Bersamina, 2026; Nieto, 2026). On a mode whose base fare f_0 covers the first d_0 kilometers and whose rate for each succeeding kilometer is r, a leg d kilometers long costs

$$f(d) = f_0 + r \cdot \lceil \max(0,\ d - d_0) \rceil \tag{9}$$

Source: fare structure and parameters reported by Daguno-Bersamina (2026) and Nieto (2026); see Table 6

where any fraction of a succeeding kilometer is charged as a whole one, a rounding convention chosen for this study. For senior citizens, students, and persons with disabilities, who receive the statutory 20 percent discount (Section 1.4.13), $f_{\delta}(d) = (1 - \delta)\, f(d)$ with $\delta = 0.20$.

$$F(J) = \sum_{\ell=1}^{n_J} f_{\mathrm{mode}(\ell)}(d_\ell) \tag{10}$$

Source: formulated in this study

Table 6. Fare parameters in effect from September 28, 2026, approved by the LTFRB and reported by Daguno-Bersamina (2026) and Nieto (2026). Tricycle fares follow local ordinances and are gathered by municipality.

Mode	Base Fare f_0 (₱)	Base Distance d_0 (km)	Rate r (₱/km)
Traditional jeepney	14.00	4	2.00
Modern jeepney	17.00	4	2.40
Provincial ordinary bus	12.00	5	2.20
Tricycle	set by ordinance	set by ordinance	set by ordinance

2.7 The Hospital Map: Capability, Services, and Cost Class

Sections 2.7 to 2.13 build the hospital layer. Nothing in Sections 2.1 to 2.6 changes: the hospital layer reads the same graphs, the same vehicle classes, and the same landmark set.

Definition 13 (Hospital record). Let $H \subseteq L_d$ be the hospitals and infirmaries in the destination set. Each $h \in H$ carries its entrance node v_h (Definition 1); its authorized bed capacity c_h; a service capability level $\eta_h \in \{0, 1, 2, 3\}$, where 0 denotes an infirmary and 1 to 3 the hospital levels of Table 7; a service set $\Sigma_h \subseteq \mathscr{S}$, where $\mathscr{S}$ is the list of services in Table 8; an emergency-room flag $e_h \in \{0, 1\}$; a cost class $q_h \in \{0, 1, 2, 3\}$; and a contact address m_h, the e-mail address that the facility publishes for official correspondence. The cost class follows the rule

$$q_h = \begin{cases} 0 & \text{if } h \text{ is government-owned and hosts a Malasakit Center,} \\ 1 & \text{if } h \text{ is government-owned without a Malasakit Center,} \\ 2 & \text{if } h \text{ is privately owned and PhilHealth-accredited,} \\ 3 & \text{if } h \text{ is privately owned and not PhilHealth-accredited.} \end{cases} \tag{11}$$

Source: this study; program flags from Republic Acts No. 11223 and 11463 (2019)

The ordering is a declared ranking of the expected out-of-pocket burden, not a price. Its inputs, namely ownership, PhilHealth accreditation, and the presence of a Malasakit Center, are all published (Appendix A), so any reader can check the class of any hospital.

The service set starts from the default list of the hospital's level, $\Sigma(\eta_h)$ in Table 8, and changes only where a published register entry for that facility says otherwise:

$$\Sigma_h = \left(\Sigma(\eta_h) \cup X_h^+\right) \setminus X_h^- \tag{12}$$

Source: formulated in this study; default lists from DOH (2012)

where X^+_h holds services the register lists beyond the level default (for example, a separately licensed dialysis unit in a Level 2 hospital) and X^-_h holds default services that the register marks as unavailable. With no such entry, $\Sigma_h = \Sigma(\eta_h)$.

Table 7. Service capability levels in the model, following the hospital classification of the DOH (2012).

η_h	Facility	Services Relevant to Routing
0	Infirmary	Treated below Level 1 and offered only to profiles whose minimum level is 0
1	Level 1 hospital	Level 1 imaging, pharmacy and blood station, secondary clinical laboratory, isolation and maternity facilities, and an operating room with recovery room
2	Level 2 hospital	What Level 1 offers, plus a general ICU, a neonatal ICU, care for high-risk pregnancy, a tertiary clinical laboratory, and Level 2 imaging
3	Level 3 hospital	What Level 2 offers, plus dialysis, a blood bank, physical medicine and rehabilitation, accredited residency training, and Level 3 imaging, including interventional radiology

Table 8. The services a patient can require in the model, the lowest level that offers each by default, and examples of patients who need them. A need is entered from a plain description or from a referral slip; the application does not diagnose.

Code	Service	Default From	Example of a Patient Who Needs It
ISO	Isolation facilities	Level 1	A patient with a contagious illness who must be kept apart from others
MAT	Maternity care (normal delivery)	Level 1	A woman in uncomplicated labor
OR	Operating room with recovery room	Level 1	A patient whose condition needs an operation

Code	Service	Default From	Example of a Patient Who Needs It
ICU	General intensive care unit	Level 2	A patient who needs close monitoring or life support
NICU	Neonatal intensive care unit	Level 2	A premature or sick newborn
HRP	Care for high-risk pregnancy	Level 2	A pregnancy with complications such as bleeding or very high blood pressure
DIA	Dialysis	Level 3	A patient with kidney failure
BB	Blood bank	Level 3	A patient who has lost a large amount of blood

Table 9. Cost classes (Equation 11) and the payment classes from which users choose. Payment class ρ admits every hospital with $q_h \le \rho$.

Class	Hospital in Cost Class q_h	Patient's Choice in Payment Class ρ	Hospitals Accepted
0	Public hospital that hosts a Malasakit Center	"Free or assisted care only"	$q_h = 0$
1	Public hospital with no Malasakit Center	"Public hospitals"	$q_h \le 1$
2	Private, PhilHealth-accredited hospital	"Private hospitals that accept PhilHealth"	$q_h \le 2$
3	Private hospital lacking accreditation	"Any hospital"	$q_h \le 3$

2.8 Agent Network for Live Bed Availability

The registers of Appendix A record authorized beds, not free ones. To learn the free count, the researchers built six kinds of software agents that write to the hospitals and ask (Table 10, Figure 2).

Table 10. Who does what in the bed-availability network.

Agent	Task
Registry	Holds the hospital records of Definition 13, the opt-out list, and the bed-status table. No other agent may write to the database.
Coordinator	Opens every inquiry cycle on the schedule of Definition 14, enforces the opt-out list and the message limits, hands inquiries to the inquiry agents, and decides whether a reminder is due.
Inquiry (one per legislative district)	Prepares the inquiry of Appendix F, attaches a one-time reference token, and sends it by SMTP to the hospital's contact address.

Agent	Task
Reader	Polls the project mailbox over IMAP every 10 minutes and once more just before any send. It picks out opt-outs, pairs each reply with its token and sender, and extracts the numbers by fixed rules. It never guesses: anything it cannot read with certainty goes to the human review queue.
Validator	Tests each reply against rules V1 to V5 of Table 11 and passes accepted reports to the Registry.
Status	On request from the application, computes bed status and free beds and reports the age of Definition 15.
Human reviewer (a person, not an agent)	A researcher who clears the review queue. A figure keyed in by hand passes through the Validator exactly like any other figure.

Definition 14 (Inquiry protocol). For hospital *h*, cycle *k* opens at

$$t_k = t_1 + (k-1)P_h,$$
$$\text{with reminders only at } t_k + jR_{rem}, \quad 1 \le j \le n_{rem}, \quad n_{rem}R_{rem} < P_h \tag{13}$$

Source: formulated in this study (Definition 14)

The default period is P_h = 8 h, one cycle per standard nursing shift. A facility that prefers a single daily inquiry is set to P_h = 24 h.

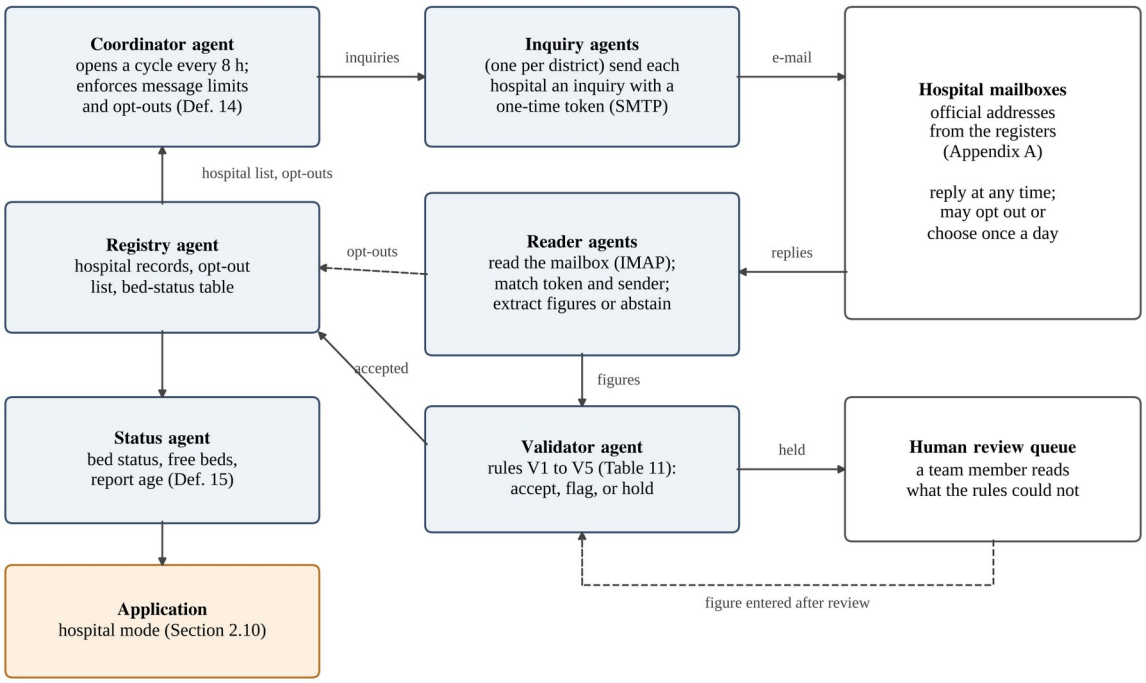


Figure 2. Structure of the bed-availability network. Solid arrows are inquiries, replies, and reports; dashed arrows are opt-outs and figures entered after human review.

Table 11. The five checks run by the Validator agent on every reply.

Rule	Name	Check	If the Check Fails
V1	Authenticity	The subject line shows a token issued to *h* less than 24 h ago, and the sender is the registered address m_h.	Discarded; nothing stored
V2	Completeness	The reply yields one capacity figure and one figure for either occupied or free beds (occupied beds are derived in the second case). Inconsistent figures count as unreadable.	Referred to human review
V3	Capacity plausibility	$0.5\ c_h \le \hat{c}_h \le 2\ c_h$, where $\hat{c}_h$ is the capacity stated in the reply.	Held, as a possible typing error or a different idea of a bed
V4	Occupancy plausibility	$0 \le \hat{o}_h \le 2\ \hat{c}_h$. A count above $\hat{c}_h$ is read as "over capacity."	Held
V5	Recency	The reply is newer than the report already on file.	Added to the history only

Rule V4 lets occupancy exceed the stated capacity. During surges, patients are often treated on stretchers in hallways or in shared spaces, and a count above capacity reflects that.

Definition 15 (Bed status and report age). For hospital *h*, a validated report is a tuple $r_h = (t^r_h, \hat{c}_h, \hat{o}_h, \hat{e}_h)$: the time of the reply, the reported capacity, the reported occupancy, and an optional emergency-room status ($\hat{e}_h \in$ {ACCEPTING, FULL}). From the most recent validated report,

$$b_h(t) = \max(0,\ \hat{c}_h - \hat{o}_h), \qquad age_h(t) = t - t^r_h \tag{14}$$

$$\sigma_h(t) = \begin{cases} \text{UNKNOWN} & \text{no validated report, or } age_h(t) > age_{\max}, \\ \text{FULL} & age_h(t) \le age_{\max} \text{ and } \big(b_h(t) = 0 \text{ or } \hat{e}_h = \text{FULL}\big), \\ \text{AVAILABLE} & \text{otherwise,} \end{cases} \tag{15}$$

Source: formulated in this study (Definition 15)

with $age_{\max}$ = 12 h. The application never shows a status without its age, for example "9 free beds, reported 1 h 40 min ago."

2.9 The Patient Profile: Acuity, Needs, Ability to Pay, and Mobility

The earlier version of hospital mode described a patient by two numbers, acuity and payment class. That leaves out two things that decide where a real patient can go: the specific service the patient needs, and how far the patient can walk to and from a stop.

Definition 16 (Patient profile). A profile is $\pi = (\kappa, N, \rho, m)$, where

- $\kappa \in \{1, \ldots, 5\}$ is the acuity level from the Emergency Severity Index, on which level 1 is the most urgent (Gilboy et al., 2011);
- $N \subseteq \mathscr{S}$ is the set of services the patient needs, from Table 8 ($N = \varnothing$ when no particular service is required);
- $\rho \in \{0, 1, 2, 3\}$ is the payment class of Table 9, with $\rho_{\max} = 3$; and
- $m \in \{\text{standard}, \text{slow}\}$ is the mobility class, which fixes the walking speed v_m and the walking cap $\bar{t}_m$ used in patient mode: $(v_{\text{standard}}, \bar{t}_{\text{standard}}) = (4.5 \text{ km/h}, 15 \text{ min})$, the values of Definition 3, and $(v_{\text{slow}}, \bar{t}_{\text{slow}}) = (3.0 \text{ km/h}, 10 \text{ min})$.

The slow class is meant for a patient who walks more slowly than the healthy adults behind Definition 3, such as an older person, a woman in late pregnancy, or a person with a disability or an injury. Its two values are declared design parameters, to be replaced when local data exist.

The acuity level is entered by the ambulance crew or chosen by the user from the plain descriptions in Table 12. The needs are chosen from the plain descriptions in Table 8 or copied from a referral slip.

Table 12. Minimum service level and emergency flag for each acuity level. Descriptions adapted from Gilboy et al. (2011); λ and K_E are declared design parameters.

κ	**Description**	$\lambda(\kappa)$	**In K_E**
1	Life-saving treatment needed at once	3	yes
2	High-risk situation; new confusion, lethargy, or disorientation; or severe pain or distress	2	yes
3	Stable; probably needs two or more hospital resources	1	no
4	Stable; probably needs one resource	1	no
5	Stable; probably needs no resources	0	no

Table 12 is a placeholder for a referral policy that the Provincial Health Office may issue. The proofs in Section 3.2 use only one property of λ: it never decreases as urgency increases, so a more urgent patient never needs a lower service level.

2.10 Finding the Nearest Suitable Hospital and Its Backups

Definition 17 (Suitable hospital). At time t, hospital h suits profile π when $\Phi(h; \pi, t) = 1$, where

$$\Phi(h;\pi,t) = \mathbf{1}[\eta_h \geq \lambda(\kappa)] \cdot \mathbf{1}[N \subseteq \Sigma_h] \cdot \mathbf{1}[\kappa \notin K_E \text{ or } e_h = 1] \\ \cdot \mathbf{1}[\sigma_h(t) \neq \text{FULL}] \cdot \mathbf{1}[\kappa \in K_E \text{ or } q_h \leq \rho] \tag{16}$$

Source: formulated in this study; service levels from DOH (2012), acuity levels from Gilboy et al. (2011), emergency exception from Republic Act No. 10932 (2017)

and $H_\pi(t) = \{\ h \in H : \Phi(h;\ \pi,\ t) = 1\ \}$ is the suitable set. The five factors check, in order, the hospital's level, its services, its emergency room, its beds, and its cost.

Three choices in Equation (16) need explaining. First, a hospital whose status is UNKNOWN stays in the suitable set.

Definition 18 (Nearest suitable hospital and backup list). For an origin o, a time t, a profile π, and a mode x, let $T_x(o, h, t)$ be the travel time from o to hospital h. In ambulance mode it is the shortest-path time from o to v_h on the ambulance subgraph $G_{\gamma amb}$ of Definition 11, with arc times ℓ_e / v_e. In patient mode a sink vertex z_h is added to the graph of Definition 5 for every hospital. The sink is reached from each arrival event at a stop within walking distance of h by an arc whose duration is the walking time to the hospital, computed with the walking speed v_m and capped at $\bar{t}_m$ of the patient's mobility class; walk arcs between stops use the same two values. Then $T_{pt}(o, h, t) = \tau^*_m(o, z_h, t) - t$. Rank the suitable hospitals by the key

$$\text{key}(h) = (T_x(o,h,t),\ -b_h(t),\ \text{id}_h), \\ h^*(o,\pi,t) = \arg\min_{h \in H_\pi(t)} \text{key}(h), \qquad T^*(o,\pi,t) = \min_{h \in H_\pi(t)} T_x(o,h,t) \tag{17}$$

Source: formulated in this study (Definition 18)

compared lexicographically. Ties in travel time go to the hospital with more free beds, and then to the smaller identifier; a hospital with UNKNOWN status counts as having zero free beds for this purpose.

The backup list answers problem P8. A hospital may become full between its last report and the patient's arrival, and an ambulance crew turned away at the door should already know where to go next and how much longer that trip takes.

Definition 19 (Stabilization rule). The rule applies only when $\kappa = 1$. Let h° be the nearest hospital with an emergency room ($e_h = 1$) whose status is not FULL, ignoring its level, services, and cost class. If such a hospital exists and

$$T_x(o, h^\circ, t) + \theta_{stab} < T^*(o, \pi, t), \tag{18}$$

Source: formulated in this study (Definition 19)

the application returns h° first, with the message "stabilize, then transfer to h^*," and h^* second. A patient who needs immediate life-saving care should not pass an open emergency room on the way to a Level 3 hospital.

Figure 3 shows the steps of hospital mode, and Algorithm 1 gives the search.

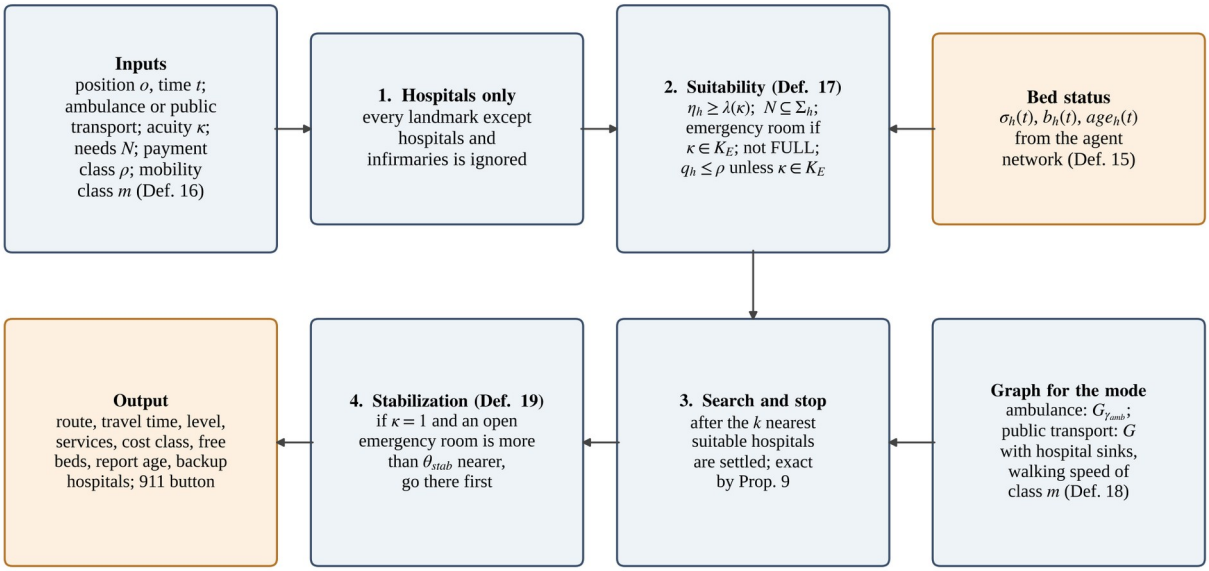


Figure 3. Steps of hospital mode, from the patient's inputs to the recommendation. Orange boxes depend on the agent network or are shown to the user.

Algorithm 1. Hospital mode for one patient (early-stopping search with backups)

```
Input : origin o, time t, profile pi = (kappa, N, rho, m), mode x, backups k
Output: up to k suitable hospitals in key order, with travel times
 1  if x = amb: graph <- ambulance subgraph, arc time = length / speed
 2  else:       graph <- time-expanded graph with hospital sinks,
                         walking speed v_m and walking cap t_m
 3  Q <- priority queue holding the origin with label t
 4  found <- empty list;  bound <- +infinity
 5  while Q is not empty:
 6      (label, u) <- extract the smallest label from Q
 7      if label > bound: stop
 8      if u is the vertex (or sink) of hospital h and Phi(h; pi, t) = 1:
 9          add (label - t, -b_h(t), id_h) to found
10          if found has k entries and bound = +infinity: bound <- label
11      relax every arc leaving u
12  sort found by key; keep the first k
13  if kappa = 1: apply the stabilization rule (Definition 19)
14  return found
```

2.11 Gaps, Catchments, and Bed Access per Person

The quantities in this section turn single routes into maps and summary figures for planners.

Definition 20 (Payment, need, and mobility gaps). At origin *o* and time *t*, for a profile π = (κ, *N*, ρ, *m*),

$$\Delta_{pay}(o,\pi,t) = T^*\big(o,(\kappa,N,\rho,m),t\big) - T^*\big(o,(\kappa,N,\rho_{\max},m),t\big) \tag{19}$$

$$\Delta_{need}(o,\pi,t) = T^*\big(o,(\kappa,N,\rho,m),t\big) - T^*\big(o,(\kappa,\emptyset,\rho,m),t\big),$$
$$\Delta_{mob}(o,\pi,t) = T^*_{pt}\big(o,(\kappa,N,\rho,\text{slow}),t\big) - T^*_{pt}\big(o,(\kappa,N,\rho,\text{standard}),t\big) \tag{20}$$

Source: formulated in this study (Definition 20)

with the convention ∞ − ∞ := 0. The payment gap is the extra travel time a patient bears because of what the family can pay.

Definition 21 (Free-bed accessibility). The free-bed analogues of Definition 7 count the free beds, and the free beds that a patient in payment class ρ may use, within budget θ of anchor *a*:

$$B_a(\theta) = \sum_{h\in H} b_h(t)\,\mathbf{1}\left[\bar{T}(o_a,h)\le\theta\right],$$
$$B^{\rho}_a(\theta) = \sum_{h\in H:\ q_h\le\rho} b_h(t)\,\mathbf{1}\left[\bar{T}(o_a,h)\le\theta\right] \tag{21}$$

Source: formulated in this study by applying Wachs and Kumagai (1973) to free beds

Definition 22 (Hospital catchment map). For a profile π and time *t* in ambulance mode, the catchment of hospital *h* is

$$C_h(\pi,t) = \{\, v\in V_R : h^*(v,\pi,t) = h \,\} \tag{22}$$

Source: formulated in this study; a network Voronoi diagram in the sense of Okabe et al. (2000) and Erwig (2000), restricted to suitable hospitals

The catchments are disjoint, and together they cover every road vertex that can reach a suitable hospital. Each catchment is the area that a given hospital would serve for that kind of patient.

Definition 23 (Bed-to-population ratio and bed desert). Following the two-step floating catchment area method of Luo and Wang (2003), let s_h be the supply of hospital *h* (its authorized beds c_h or its free beds $b_h(t)$) and, for budget θ,

$$R_h(\theta) = \frac{s_h}{\sum_{a:\, \bar{T}(o_a,h) \le \theta} P_a}, \qquad A_a^F(\theta) = \sum_{h:\, \bar{T}(o_a,h) \le \theta} R_h(\theta) \tag{23}$$

Source: Luo and Wang (2003), applied in this study to anchors and to free beds

with $R_h(\theta) = 0$ when no anchor lies within θ of h. R_h is the number of beds per person in the hospital's reach, and A^F_a is the number of beds per person that anchor a can draw on. Results are reported per 10,000 people.

The difference between Equations (4) and (23) matters to patients. In Equation (4), one bed counts once for every anchor that can reach it, so a bed shared by fifty thousand people looks the same as a bed shared by five hundred.

Algorithm 2 builds the whole catchment map in one search, and Proposition 14 proves it correct.

Algorithm 2. Catchment map for one profile from a single search

```
Input : ambulance subgraph, profile pi, time t
Output: for every road vertex v: h*(v), T*(v); shortest-path forest F
 1  reverse every arc of the ambulance subgraph
 2  for each hospital h with Phi(h; pi, t) = 1:
 3      insert v_h into Q with label (0, -b_h(t), id_h) and no parent
 4  while Q is not empty:
 5      ((d, beta, i), u, p) <- extract the smallest label (lexicographic)
 6      if u is already labeled: continue
 7      label(u) <- (d, beta, i);  parent(u) <- p
 8      for each reversed arc (u, w) with travel time delta:
 9          if w is unlabeled: push w, label (d + delta, beta, i), parent u
10  F <- { road segment between u and parent(u) : parent(u) exists }
11  return labels and F      (hospital i is h*(u); d is T*(u))
```

2.12 Surge Assignment: Many Patients, Few Beds

Hospital mode answers one patient at a time. During a typhoon, an eruption, or a large road crash, many patients need beds at once, and the free beds are shared.

Definition 24 (Surge assignment). Let patients $i = 1, \ldots, n$ wait at origins o_i with profiles π_i, and let each hospital h have $b_h \ge 0$ free beds. Let $\Phi'(h; \pi_i, t)$ be Equation (16) without the FULL factor, since capacity is now handled by b_h directly, and let $c_{ih} = T_x(o_i, h, t)$. The surge assignment chooses $x_{ih} \in \{0, 1\}$ to

$$\begin{aligned} \text{minimize} \quad & \sum_{i}\sum_{h} c_{ih}\,x_{ih} + \sum_{i} U_i\Big(1 - \sum_{h} x_{ih}\Big) \\ \text{subject to} \quad & \sum_{h} x_{ih} \le 1 \ \text{ for every } i, \qquad \sum_{i} x_{ih} \le b_h \ \text{ for every } h, \\ & x_{ih} = 0 \ \text{ whenever } \Phi'(h;\pi_i,t) = 0 \text{ or } c_{ih} = +\infty, \end{aligned} \tag{24}$$

Source: formulated in this study as a transportation problem (Ahuja et al., 1993)

where U_i is the penalty for leaving patient i unplaced. With $\hat{c}$ the largest finite c_{ih}, the penalties are set to $U_i = U_0 > n\,\hat{c}$ for non-emergency patients and $U_i = U_E > n\,(U_0 + \hat{c})$ for patients with $\kappa_i \in K_E$.

Algorithm 3. Surge assignment

```
Input : patients i with origins o_i and profiles pi_i; free beds b_h
Output: a hospital, or "not placed", for each patient
 1  for each patient i and hospital h:
 2      cost(i, h) <- T_x(o_i, h, t) if h suits pi_i (ignoring FULL)
                      and is reachable; otherwise forbidden
 3  make b_h copies ("bed slots") of each hospital h
 4  give each patient i one private "not placed" slot with cost U_i
 5  solve the assignment problem on patients x slots
        (Hungarian method, Kuhn, 1955, or minimum-cost flow)
 6  return the hospital of each patient's slot, or "not placed"
```

2.13 Planning Measures: Critical Roads and Where to Add Care

Definition 25 (Segment criticality). For a profile π and time t, let $G - e$ be the ambulance subgraph with road segment e closed in both directions. The criticality of e is

$$D_e(\pi,t) = \sum_{a \in L_a} P_a \left[T^{*}_{G-e}(o_a,\pi,t) - T^{*}_{G}(o_a,\pi,t) \right] \tag{25}$$

Source: formulated in this study, after the link-importance measure of Jenelius et al. (2006)

with $\infty - \infty := 0$, and $D_e = +\infty$ when closing e cuts some anchor off from every suitable hospital. Segments with large D_e are the ones to clear first after a typhoon and to protect before one.

Definition 26 (Site selection by coverage). Fix a budget θ and a profile for a patient in payment class 0, for example $\kappa = 3$ and $N = \varnothing$. Let C_0 be the anchors already within θ of a suitable hospital of cost class 0, let Q be the candidate sites (for example, public hospitals of cost class 1 that could host a new Malasakit Center), and let $C_j = \{\, a : \bar{T}(o_a, j) \le \theta \,\}$ for $j \in Q$. The coverage gained by a set of sites $S \subseteq Q$ is

$$f(S) = \sum_{a \in U(S) \setminus C_0} P_a, \qquad U(S) = C_0 \cup \bigcup_{j \in S} C_j \tag{26}$$

Source: formulated in this study after the maximal covering location problem of Church and ReVelle (1974)

and the task is to choose $|S| = k$ sites with the largest $f(S)$. Algorithm 4 does this greedily, and Proposition 18 bounds how far it can fall short.

Algorithm 4. Greedy site selection

```
Input : anchors a with weights P_a; candidate sites Q; covered set C_0;
        coverage sets C_j; number of sites k
Output: chosen sites S
 1  S <- empty set;  U <- C_0
 2  repeat k times:
 3      for each j in Q not in S:  gain(j) <- sum of P_a over a in C_j, a not in U
 4      j* <- the j with the largest gain (smallest index on ties)
 5      S <- S plus j*;  U <- U plus C_j*
 6  return S
```

2.14 Assumptions

Shortest-path searches on time-dependent networks keep their optimality only if the first-in, first-out property holds (Kaufman & Smith, 1993; Pyrga et al., 2008). The first two assumptions secure it; the others fix the scope of the model.

- **Assumption 1 (Nonnegative traversal).** A journey never goes back in time, because no arc of G has $t(\text{head}) < t(\text{tail})$.
- **Assumption 2 (No overtaking within a route).** On any given route, a vehicle that leaves earlier does not arrive later than one that leaves after it.
- **Assumption 3 (Frequency of service).** On routes with no fixed schedule, departures are placed at the average headway. The sensitivity tests compare the resulting waits with a half-headway baseline.
- **Assumption 4 (No queue at the destination).** On arrival at a landmark, no one waits in line or meets a capacity limit. This is standard in cumulative accessibility studies (Geurs & van Wee, 2004). Hospital mode relaxes it through live bed reports, and the surge assignment relaxes it further for patients who travel at the same time.
- **Assumption 5 (Open data only).** Every number in this paper comes from one of four sources: a mathematical derivation, a declared model parameter, a public register

(Appendix A), or a synthetic data set made only for testing. No facility was interviewed and no field measurement was taken.

- **Assumption 6 (Reported bed counts).** In live operation, bed counts are entered by the hospitals themselves. The Validator checks their form and range (Table 11) but cannot confirm them.
- **Assumption 7 (Emergency care).** Hospitals with an emergency room follow Republic Act No. 10932 (2017) and give basic emergency care in emergency or serious cases without first asking for a deposit or advance payment. The payment filter is therefore dropped for $\kappa \in K_E$.
- **Assumption 8 (Ambulance travel time).** Ambulance times use the free-flow speeds of Definition 2. Dispatch time, handover time, emergency driving, and traffic delay are not included.
- **Assumption 9 (Service lists).** A hospital offers the services of its level unless a published register entry for that facility says otherwise (Equation 12).
- **Assumption 10 (Surge requests).** In the surge assignment, the free beds are known when the assignment is made and the patients' requests are handled together, in one batch.

2.15 Notation

Table 13. Symbols and their meanings.

Symbol	Meaning
L, L_a, L_d	all landmarks; anchors; essential destinations
N_a; P_a, W	number of anchors; weight of anchor a; sum of all weights
S, C	stops; connections
$G = (V, E)$	the time-expanded timetable graph
θ_{ch}	minimum change time between trips
$\tau^*(o, d, t_0)$; T, $\bar{T}$	earliest arrival; door-to-door travel time; its mean over departure times
θ	travel-time budget
$A^k_a(\theta)$	total size of category-k destinations reachable from anchor a within θ
μ; $G(A, P)$, $T(A, P)$	weighted mean access; Gini coefficient; Theil index
γ, E_γ, d_γ; γ_{amb}	vehicle class, arcs open to it, its shortest-path distance; the ambulance class

Symbol	Meaning
$f(d)$, $F(J)$	fare for a leg; fare for a journey
H, v_h, c_h	hospitals; entrance node; authorized beds
η_h, Σ_h, e_h, q_h, m_h	service level; service set; emergency-room flag; cost class; contact address
$\mathscr{S}$; $\Sigma(\eta)$	list of services (Table 8); default service set of level η
P_h, R_{rem}, n_{rem}, D_{rep}	inquiry period; reminder delay; reminders allowed per cycle; longest reply delay assumed
$\hat{c}_h$, $\hat{o}_h$, $\hat{e}_h$, $b_h(t)$	reported capacity, occupied beds, and emergency-room status; free beds
$\sigma_h(t)$, $age_h(t)$, $age_{\max}$	bed status; age of the report; limit on age
$\pi = (\kappa, N, \rho, m)$	patient profile: acuity, needed services, payment class, mobility class
$\lambda(\kappa)$, K_E	lowest acceptable service level; emergency set
v_m, $\bar{t}_m$	walking speed and walking cap of mobility class m
Φ, $H_\pi(t)$	suitability indicator; set of suitable hospitals
T_x, h^*, T^*; $h_{(1)}$, …, $h_{(k)}$	travel time in mode x; nearest suitable hospital; time to reach it; backup list
$h°$, θ_{stab}	nearest open emergency room; margin for stabilization
Δ_{pay}, Δ_{need}, Δ_{mob}	payment gap; need gap; mobility gap
B_a, B^ρ_a	free beds reachable; free beds reachable that payment class ρ accepts
$C_h(\pi, t)$	catchment of hospital h
R_h, A^F_a	beds per person in the reach of h; beds per person available to anchor a
x_{ih}, U_i	assignment of patient i to hospital h; penalty for leaving i unplaced
D_e	criticality of road segment e
Q, C_j, C_0, $f(S)$	candidate sites; anchors covered by site j; anchors already covered; coverage gained by sites S

Sources of the equations. Equations (2) to (7) and (9) are established constructions, cited at first use, as are the Theil decomposition (27) and the bound (32) in Section 3.2. Equations (21) and (23) apply Wachs and Kumagai (1973) and Luo and Wang (2003) to anchors and free beds.

3. FINDINGS

3.0 Note on Data Sources and Integrity

Every result in this chapter comes from proof or from simulation.

Two scripts produced the numbers.

3.1 General Methods

The work proceeded in nine stages.

Stage 1: Road network. OpenStreetMap road data were used to build the network of Definition 2 (OpenStreetMap contributors, 2026).

Stage 2: Landmark listing and snapping. The public registers of Table 4 supplied candidate landmarks under the criteria of Appendix C.

Stage 3: GTFS feed. Stops, routes, trips, arrival times, calendar rules, fare rules, and route shapes were written into a standard GTFS feed (Appendix B).

Stage 4: Graph and queries. The feed and the walking links were combined into the time-expanded graph of Section 2.2.

Stage 5: Accessibility and inequality. Cumulative landmark scores were computed from the travel times, and their distributions gave the Lorenz curves, Gini coefficients, Theil indices, and Palma ratios.

Stage 6: Hospital map. Every hospital and infirmary received the record of Definition 13.

Stage 7: Live bed tracking. The e-mail agents of Section 2.8 followed the protocol of Definition 14 and published free beds, status, and report age for each facility (Definition 15).

Stage 8: Hospital mode. The routing engine, the hospital map, and the bed table were joined into hospital mode (Algorithm 1), with the suitability rule of Definition 17, the backup list of Definition 18, and the stabilization rule of Definition 19.

Stage 9: Patient needs and planning tools. The patient profile was extended with required services and mobility classes (Definition 16).

The web application that joins these parts, with its fare calculator, route directions, vehicle-restricted routing, and hospital mode, is described in Appendix D.

3.2 Theoretical Results

The query engine must be shown correct before any accessibility figure built on it can be trusted, so the results begin there.

Proposition 1 (Query engine correctness; from Dijkstra, 1959; Pyrga et al., 2008). Suppose Assumption 1 holds. For any origin o, destination d, and departure time t_0, a label-setting shortest-path search on the time-expanded graph G that uses absolute arrival time as its label and extracts vertices in nondecreasing order of label returns the exact earliest arrival $\tau^*(o, d, t_0)$.

Proof. Weight each arc (u, v) of E by $\omega(u, v) = t(v) - t(u)$; under Assumption 1 every such weight is nonnegative. Along any path π that leaves the source (o, t_0) and ends at (d, t), the weights telescope to $\Sigma_{(u,v) \in \pi}\ \omega(u, v) = t - t_0$, so a path has minimum weight exactly when it arrives earliest. By the way E is built, paths in G and feasible journeys are the same objects: a ride arc never lets the rider off between two consecutive stops, a change of trip always passes through a transfer arc costing θ_{ch}, a wait arc lets the rider stay put, and no walk arc is longer than $\bar{t}_W$. A minimum-weight path is therefore a feasible journey with the earliest possible arrival. Nonnegative weights make Dijkstra's (1959) algorithm label-setting (Cormen et al., 2022). Whenever the vertex with the smallest tentative label is taken from the queue, nothing still queued carries a smaller label, and since no arc weight is negative, no path found later can improve it. Each label is final once its vertex is removed, and the first vertex removed at stop d carries the label $\tau^*(o, d, t_0)$. □

Remark 1 (Why the vertex set must be event-based). Proposition 1 depends on the event-based vertices of Definition 5, one for each departure and each arrival. A common shortcut keeps a single label per stop, and that breaks the result.

Proposition 2 (Computational complexity; Cormen et al., 2022). Let $|S|$ be the number of stops, $|C|$ the number of connections, and $\bar{f}$ the mean number of footpaths per stop. Then $|V| = O(|C| + |S|)$ and $|E| = O((1 + \bar{f})|C| + |S|)$. The search of Proposition 1 runs in $O(|E| \log |V|)$ per query with a binary heap. If every arc duration is strictly positive, G is acyclic and the same query can be answered in $O(|V| + |E|)$ by a scan in topological order.

Proof. Each connection contributes one departure vertex, one arrival vertex, and one ride arc. Wait arcs form a chain through the events at each stop, so they add $O(1)$ arcs per event. Transfer and walk arcs add $O(1 + \bar{f})$ arcs per arrival event. The heap bound is the standard bound for Dijkstra's algorithm with nonnegative weights. In the acyclic case, every arc strictly increases the time coordinate, so no directed cycle exists, and relaxing the arcs in topological order visits each arc once. □

Proposition 3 (Nested vehicle limits and monotone reachability). If $\gamma \le \gamma'$, then $E_{\gamma'} \subseteq E_{\gamma}$, and the shortest-path times satisfy $d_{\gamma}(u, v) \le d_{\gamma'}(u, v)$ for all $u, v \in V_R$. Every vertex reachable for γ' is reachable for γ.

Proof. Take any arc $e \in E_{\gamma'}$. For each component i, $a^i_e \ge \gamma'^i \ge \gamma^i$, so $e \in E_{\gamma}$ and therefore $E_{\gamma'} \subseteq E_{\gamma}$. Every path of $G_{\gamma'}$ uses only arcs of $E_{\gamma'}$, so it is also a path of G_{γ}. A minimum over a set of paths cannot exceed the minimum over a subset of it, which gives the distance inequality, and comparing finite path lengths gives the reachability claim. The argument needs only nonnegative arc weights, so it holds for the travel-time weights ℓ_e / v_e. □

Corollary 4 (Vehicle-class accessibility ordering). For every anchor a, category k, and budget θ, if $\gamma \le \gamma'$ then $A^k_{a,\gamma}(\theta) \ge A^k_{a,\gamma'}(\theta)$. A larger vehicle class never has more access than a smaller one.

Proof. With travel times as weights, Proposition 3 makes every travel time under γ' at least the matching time under γ. The indicator $\mathbf{1}[\bar{T} \le \theta]$ does not increase as $\bar{T}$ grows, and a weighted sum with nonnegative weights w_p inherits that property. □

Remark 2 (Preprocessing many vehicle classes). Corollary 4 also makes multi-class routing cheap. The classes form a partially ordered set, not a single chain; the tricycle needs more clearance than the sedan, while the sedan needs more width.

Proposition 5 (Standard properties of the access Gini coefficient; Cowell, 2011). Let $A_a \ge 0$ for every anchor, with weights $P_a > 0$ and weighted mean $\mu > 0$. Then (i) $0 \le G(A, P) \le 1$; (ii) $G(\lambda A, P) = G(A, P)$ for every $\lambda > 0$; and (iii) $G(A + c\,\mathbf{1}, P) < G(A, P)$ for every $c > 0$ whenever $G(A, P) > 0$.

Proof. (i) Every term of the numerator is nonnegative, so $G \ge 0$. Since $A_a \ge 0$, $|A_i - A_j| \le A_i + A_j$, hence $\Sigma_i \Sigma_j P_i P_j |A_i - A_j| \le \Sigma_i \Sigma_j P_i P_j (A_i + A_j) = 2W \Sigma_i P_i A_i = 2W^2 \mu$, and dividing by $2W^2\mu$ gives $G \le 1$. (ii) Multiplying A by λ multiplies every difference and the mean by λ, and the factors cancel. (iii) Adding c to every score leaves every pairwise difference unchanged and raises the mean from μ to $\mu + c$, so a positive G strictly falls. □

Remark 3 (What the Gini properties mean for planners). Property (ii) allows Gini values to be compared across destination types measured in different units, such as beds and classrooms,

because the units cancel. Property (iii) gives a plain policy reading: an equal absolute increase in access for every anchor lowers measured inequality.

Proposition 6 (Exact decomposition of the Theil index; Theil, 1967; Shorrocks, 1980). Partition the anchors into groups g (here, municipalities). Let $W_g = \Sigma_{a \in g} P_a$, let μ_g be the weighted mean of group g, and let $s_g = W_g \mu_g / (W \mu)$ be the group's share of all reachable destinations. Then

$$T = \sum_g s_g T_g + \sum_g s_g \ln \frac{\mu_g}{\mu} \tag{27}$$

Source: Theil (1967); decomposition after Shorrocks (1980)

where T_g is the Theil index within group g.

Proof. Write $p_a = P_a / W$ and, for $a \in g$, $p^g_a = P_a / W_g$, so $p_a = (W_g / W)\, p^g_a$. Split the ratio as $A_a / \mu = (A_a / \mu_g)(\mu_g / \mu)$. Then

$$T = \sum_g s_g \left[\sum_{a \in g} p^g_a \frac{A_a}{\mu_g} \ln \frac{A_a}{\mu_g} + \ln \frac{\mu_g}{\mu} \sum_{a \in g} p^g_a \frac{A_a}{\mu_g} \right] \tag{28}$$

Source: expansion of Equation (27); Theil (1967), Shorrocks (1980)

Inside the brackets, the first sum is T_g by definition, and the second is $\Sigma_{a \in g} P_a A_a / (W_g \mu_g) = 1$.

Remark 4 (From one number to a diagnosis). With Proposition 6, a single inequality figure becomes a diagnosis. A large within-municipality part points to a local problem, to be met with feeder routes, better stop placement, or a closer health facility inside the municipality.

Remark 5 (Decomposition over a chosen index set). Q7 asked whether the decomposition survives when the index set is a list of landmarks rather than a full partition of the province. It does.

Proposition 7 (Monotonicity and subadditivity of regulated fares). The leg fare $f(d) = f_0 + r \lceil \max(0, d - d_0) \rceil$ never decreases as d grows. If, in addition, $r\, d_0 \le f_0$ and d_0 is a positive integer, then f is subadditive: $f(d_1 + d_2) \le f(d_1) + f(d_2)$ for all $d_1, d_2 \ge 0$.

Proof. Monotonicity is immediate, since $d \mapsto \lceil \max(0, d - d_0) \rceil$ never decreases and $r > 0$. For subadditivity, put $x = \max(0, d_1 - d_0)$ and $y = \max(0, d_2 - d_0)$. From $d_1 \le x + d_0$ and $d_2 \le y + d_0$ it

follows that $\max(0, d_1 + d_2 - d_0) \le x + y + d_0$. The ceiling of a sum never exceeds the sum of the ceilings, and the integer d_0 is its own ceiling, so $\lceil \max(0, d_1 + d_2 - d_0) \rceil \le \lceil x \rceil + \lceil y \rceil + d_0$. Hence

$$\begin{aligned} f(d_1 + d_2) &\le f_0 + r\lceil x \rceil + r\lceil y \rceil + r d_0 \\ &\le 2f_0 + r\lceil x \rceil + r\lceil y \rceil = f(d_1) + f(d_2) \end{aligned} \tag{29}$$

Source: formulated in this study (Proposition 7)

with $r\, d_0 \le f_0$ giving the second step. □

Corollary 8 (Fare against time). For the traditional jeepney in Table 6, $r\, d_0 = 8 \le 14 = f_0$, so its fare is subadditive: one ride over a distance never costs more than two rides that cover the same distance. The cheapest journey therefore prefers a few long legs, while the quickest may use more transfers. Because the two aims can pull apart, the planner offers the whole Pareto set over arrival time, fare, and number of transfers rather than a single journey. For a poor patient going to hospital by public transport, the cheapest option is shown beside the fastest.

Remark 6 (Why the Pareto set stays small). Shortest paths under two or more additive criteria are hard in general. The weight-constrained version is already NP-complete (Garey & Johnson, 1979), and the number of Pareto-optimal paths can grow exponentially with the size of the network (P.

The remaining results concern the hospital layer.

Proposition 9 (Early stopping with backups loses no exactness). Fix o, t, π, and $k \ge 1$. Search from o at time t by the label-setting method of Proposition 1 in mode *pt* (with the walking values of the patient's mobility class) or by Dijkstra's algorithm on $G_{\gamma amb}$ in mode *amb*. Halt once the k-th vertex of a suitable hospital has been extracted, together with every other vertex that shares its label. Then the suitable hospitals extracted, sorted by the key of Equation (17), begin with exactly the backup list $(h_{(1)}, \ldots, h_{(k)})$ of Definition 18, and the label of each, minus t, equals its travel time $T_x(o, h_{(j)}, t)$. For $k = 1$ this gives $h^*(o, \pi, t)$ and $T^*(o, \pi, t)$. If fewer than k suitable hospitals can be reached, the queue empties and the list holds all of them; if none can be reached, $T^* = +\infty$.

Proof. Both graphs have nonnegative arc weights (G by Assumption 1, $G_{\gamma amb}$ because lengths and speeds are positive), so labels leave the queue in nondecreasing order and are final when they leave, exactly as in Proposition 1. Let L be the label at which the k-th suitable hospital vertex

leaves the queue. Every suitable hospital with a label below L left the queue earlier, because labels leave in order. Every suitable hospital with label exactly L leaves before the search halts, by the halting rule. Every other suitable hospital has a label above L. At least k suitable hospitals therefore have labels at most L, and since the first component of the key is the travel time, the first k hospitals in key order all have labels at most L and were all extracted with exact labels. Sorting the extracted hospitals by key therefore yields $h_{(1)}, \ldots, h_{(k)}$. In patient mode the sink z_h is entered only through walking arcs from arrival events, so its label is the earliest possible arrival at the hospital and the travel time is $L_h - t$. If fewer than k suitable hospitals are reachable, the halting rule never fires, the queue empties, and every reachable suitable hospital has been extracted with its exact label. □

Remark 7 (Why hospital mode is fast). An atlas query must find paths from an origin to every landmark. A hospital query needs only the few nearest suitable hospitals, so it can stop early.

Proposition 10 (Suitability is monotone in ability to pay, acuity, and needs). Fix o, t, the mode of travel, the mobility class, and every bed status.

(i) If $\rho \le \rho'$, then $H_{(\kappa, N, \rho, m)}(t) \subseteq H_{(\kappa, N, \rho', m)}(t)$ and $T^*(o, (\kappa, N, \rho', m), t) \le T^*(o, (\kappa, N, \rho, m), t)$.

(ii) If κ and κ' are both inside K_E or both outside it, and $\lambda(\kappa) \ge \lambda(\kappa')$, then $H_{(\kappa, N, \rho, m)}(t) \subseteq H_{(\kappa', N, \rho, m)}(t)$ and $T^*(o, (\kappa', N, \rho, m), t) \le T^*(o, (\kappa, N, \rho, m), t)$.

(iii) If $\kappa \in K_E$, then $H_{(\kappa, N, \rho, m)}(t) = H_{(\kappa, N, \rho\max, m)}(t)$ whatever ρ is.

(iv) If $N \subseteq N'$, then $H_{(\kappa, N', \rho, m)}(t) \subseteq H_{(\kappa, N, \rho, m)}(t)$ and $T^*(o, (\kappa, N, \rho, m), t) \le T^*(o, (\kappa, N', \rho, m), t)$.

Proof. The travel time $T_x(o, h, t)$ depends on the mode and the mobility class but not on κ, N, or ρ, so in every part only the suitable set changes, and a minimum over a set is never below the minimum over a set that contains it. (i) Of the five factors in Equation (16), only the last involves ρ, and $\mathbf{1}[\kappa \in K_E \text{ or } q_h \le \rho]$ does not decrease as ρ grows, so the suitable sets nest. (ii) When κ and κ' fall on the same side of K_E, the third and fifth factors agree for both profiles and the fourth ignores κ, while $\lambda(\kappa) \ge \lambda(\kappa')$ gives $\mathbf{1}[\eta_h \ge \lambda(\kappa)] \le \mathbf{1}[\eta_h \ge \lambda(\kappa')]$. (iii) With $\kappa \in K_E$, the fifth factor equals 1 whatever ρ is. (iv) Only the second factor involves the needs, and $N' \subseteq \Sigma_h$ implies $N \subseteq \Sigma_h$, so the factor for N' never exceeds the factor for N. □

Remark 8 (A deliberate break at the emergency boundary). Pairs of profiles on opposite sides of K_E are left out of Proposition 10(ii) on purpose. Moving from $\kappa = 3$ to $\kappa = 2$ raises the level requirement, which removes hospitals, but it also switches off the payment filter, which adds them back.

Corollary 11 (Payment gap and need gap). For all o, t, and π: (a) $\Delta_{pay} \geq 0$, Δ_{pay} does not increase as ρ grows, $\Delta_{pay} = 0$ at $\rho = \rho_{\max}$, and $\Delta_{pay} = 0$ for every $\kappa \in K_E$; (b) $\Delta_{need} \geq 0$, $\Delta_{need} = 0$ when $N = \varnothing$, and $N \subseteq N'$ implies $\Delta_{need}(N) \leq \Delta_{need}(N')$.

Proof. (a) Proposition 10(i) gives the first three statements and Proposition 10(iii) the last. If T^* at $\rho_{\max}$ is infinite, then not even dropping the payment filter makes a suitable hospital reachable; T^* at ρ is infinite too, and the convention of Equation (19) sets $\Delta_{pay} = 0$. If only T^* at ρ is infinite, $\Delta_{pay} = +\infty$, a case reported on its own as "no affordable suitable hospital reachable." (b) Apply Proposition 10(iv) with $\varnothing \subseteq N \subseteq N'$; the subtrahend in Equation (20) is the same for N and N'. □

Remark 9 (How to read the gaps and the bed vectors). Outside emergencies, the payment gap counts the minutes that a family's budget adds to the trip; for emergency patients it is zero by construction. The need gap counts the minutes that a particular medical need adds.

Remark 10 (The ambulance's place in the vehicle order). In every component of Table 5, the ambulance lies between the sport utility vehicle and the jeepney. Proposition 3 therefore opens to the ambulance every road that a jeepney-class vehicle may use, and, with equal arc speeds, Corollary 4 guarantees that its hospital access is at least a jeepney's.

Proposition 12 (Bounds on messages, opt-outs, and report age). Assume the protocol of Definition 14. (i) In each cycle no hospital receives more than $1 + n_{rem}$ messages, and in each day no more than $(1 + n_{rem}) \lceil 24\ \text{h} / P_h \rceil$. (ii) Nothing is sent to a hospital once its opt-out is in the project mailbox. (iii) A hospital that answers each inquiry within D_{rep}, with a reply that passes validation, has $age_h(t) < P_h + D_{rep}$ at all times after its first such reply, and so it never becomes UNKNOWN if $P_h + D_{rep} \leq age_{\max}$.

Proof. (i) Within a cycle the Coordinator sends one inquiry at t_k and reminders only at $t_k + j\,R_{rem}$ with $j \leq n_{rem}$, and the condition $n_{rem}\,R_{rem} < P_h$ keeps every reminder inside its own cycle. (ii) The Reader clears the inbox immediately before each send, and the Coordinator reads the opt-out list only after that, so an opt-out already in the mailbox when a send falls due is recorded before the

send. (iii) Write $d_k \le D_{rep}$ for the time the hospital took to answer cycle k. For $t \ge t_1 + d_1$, choose the largest k with $t_k + d_k \le t$. Then $t < t_{k+1} + d_{k+1} \le t_k + P_h + D_{rep}$, and since rule V5 keeps the most recent report, the report on file dates from $t_k + d_k$ or later. Thus

$$age_h(t) \le t - (t_k + d_k) < P_h + D_{rep} - d_k \le P_h + D_{rep} \tag{30}$$

Source: formulated in this study (Proposition 12)

□

With the defaults of Definition 14 (P_h = 8 h, n_{rem} = 1) and replies within D_{rep} = 2 h, Proposition 12 guarantees at most 2 messages per cycle and 6 per day, and a report age below 10 hours, well inside the 12-hour limit of Definition 15.

Remark 11 (What Proposition 12 does not promise). Proposition 12 bounds how many messages are sent and how old the reports of responsive hospitals can get. It promises nothing about whether hospitals answer, which stays entirely voluntary.

Proposition 13 (Slower walking never shortens a trip). Let mobility classes m and m' satisfy $v_m \le v_{m'}$ and $\bar{t}_m \le \bar{t}_{m'}$. Then for every origin o, every destination d (a stop or a hospital sink z_h), and every departure time t, $\tau^*_m(o, d, t) \ge \tau^*_{m'}(o, d, t)$. Consequently, for every profile, T^*_{pt} for the slow class is at least T^*_{pt} for the standard class, $\Delta_{mob} \ge 0$, and every hospital that a slow walker can reach is reachable by a standard walker.

Proof. Take any journey j that is feasible for class m. It is a sequence of rides, waits, transfers, and walking legs. Build a journey j' for class m' leg by leg, keeping the same rides. A walking leg of length ℓ takes $\ell / v_{m'} \le \ell / v_m \le \bar{t}_m \le \bar{t}_{m'}$, so it is allowed for m' and ends no later than the same leg of j. If the leg ends at a stop, the walk arc of m' lands on the earliest departure event at or after its arrival, which is no later than the departure event that j boards there, and the wait arcs of Definition 5, which join consecutive departure events at a stop, carry j' forward to that same event. If the leg ends at the hospital, j' arrives no later than j. Rides, waits, and transfers are copied unchanged. Hence j' is feasible for m' and arrives no later than j, and taking the minimum over all j gives the inequality. Suitability does not depend on the mobility class, so both classes face the same suitable set, and a minimum of pointwise larger travel times over the same set is larger. □

Proposition 14 (The whole catchment map from one search). In ambulance mode, Algorithm 2 labels every road vertex v with $(T^*(v, \pi, t), -b_{h*}(t), \mathrm{id}_{h*})$, where $h^* = h^*(v, \pi, t)$; that is, it computes the nearest suitable hospital and its travel time for every vertex at once, with the tie-break of Definition 18, in $O(|E_{\gamma amb}| \log |V_R|)$ time, the cost of a single ordinary search. The parent pointers form a forest F of shortest paths, one tree for each suitable hospital, and the tree containing v is rooted at $v_{h*(v)}$.

Proof. For each vertex v, define the target label $\Lambda(v) = \min_{h \in H\pi(t)} (d(v, v_h), -b_h(t), \mathrm{id}_h)$ in lexicographic order, where d is the forward shortest-path time on $G_{\gamma amb}$. By Definition 18, the hospital named in $\Lambda(v)$ is $h^*(v)$ and its first component is $T^*(v)$. Reversing every arc turns $d(v, v_h)$ into the shortest-path time from v_h to v, so $\Lambda(v)$ is the smallest label over paths that start at a seeded hospital vertex in the reversed graph. Labels are triples compared lexicographically, and an arc adds its nonnegative travel time to the first component only. Hence extending a path never makes its label smaller, and two labels keep their order when the same arc extends both. These are the only two facts that the correctness argument for Dijkstra's algorithm uses (Cormen et al., 2022), so every vertex leaves the queue carrying $\Lambda(v)$. Each vertex is labeled once and each arc relaxed once, which gives the heap bound. The parent of v carries the same hospital index as v and lies on a shortest path from v to that hospital, so following parents from v ends at $v_{h*(v)}$. □

Proposition 15 (Bed-to-population ratios never count a bed twice). Let H_θ be the hospitals that have at least one anchor within θ. Then

$$\begin{aligned}\sum_a P_a A_a^F(\theta) &= \sum_a P_a \sum_{h:\, \bar{T}(o_a,h)\le\theta} R_h(\theta) \\ &= \sum_h R_h(\theta) \sum_{a:\, \bar{T}(o_a,h)\le\theta} P_a = \sum_{h\in H_\theta} s_h\end{aligned} \tag{31}$$

Source: formulated in this study (Proposition 15), for the measure of Luo and Wang (2003)

so the weighted mean of A^F equals the total supply within reach of the population divided by W.

Proof. The second equality exchanges the order of summation over the finite set of anchor-hospital pairs with $\bar{T}(o_a, h) \le \theta$. For $h \in H_\theta$ the inner sum is the denominator of $R_h(\theta)$ in Equation (23), so $R_h(\theta)$ times it equals s_h; for $h \notin H_\theta$ the inner sum is empty. □

By contrast, the cumulative measure gives $\Sigma_a\, P_a\, A_a(\theta) = \Sigma_h\, s_h\, \Sigma_{a:\, \bar{T}(oa,h) \le \theta}\, P_a$, which counts each bed once for every person who can reach it.

Proposition 16 (Optimal surge assignment). Consider program (24).

(i) Its linear relaxation has an optimal solution in whole numbers, and Algorithm 3 finds an optimal assignment in polynomial time, $O((n + B)^3)$ with the Hungarian method, where $B = \Sigma_h\, b_h$ (Kuhn, 1955; Ahuja et al., 1993).

(ii) Among all feasible assignments, an optimal one places the largest possible number of emergency patients; among those, the largest possible number of patients in all; and among those, it has the least total travel time.

(iii) If the independent choices, in which every patient goes to his or her own nearest suitable hospital with $b_h > 0$, never exceed any hospital's free beds, then they form an optimal assignment.

Proof. (i) After adding one slack variable per patient for "not placed," each column of the constraint matrix has at most one 1 among the patient rows and at most one 1 among the hospital rows, so the matrix is the incidence matrix of a bipartite graph and is totally unimodular. A linear program with such a matrix and integer right-hand sides has an integral optimal vertex (Ahuja et al., 1993). Expanding each hospital into b_h bed slots turns the program into an assignment problem with $n + B$ rows at most, which the Hungarian method solves in the stated time (Kuhn, 1955). The capacity constraint is part of the program, so no hospital is overfilled.

(ii) Let x and y be feasible, and let x place e_x emergency patients and r_x others, and y place e_y and r_y.

(iii) Each independent choice is the patient's own minimum c_{ih} over the suitable hospitals with a free bed.

Proposition 17 (Only roads on the shortest-path forest matter). Fix a profile π and time t, and let F be the forest of Proposition 14. (i) $D_e(\pi, t) \ge 0$ for every road segment e. (ii) If e is not in F, closing e changes neither h^* nor T^* at any road vertex, and $D_e(\pi, t) = 0$. Hence only the segments of F, at most $|V_R| - 1$ of them, need to be tested, each with one run of Algorithm 2.

Proof. (i) $G - e$ is a subgraph of G, so by the argument of Proposition 3 no shortest-path time falls when e is closed, and suitability does not depend on the roads. (ii) If $e \notin F$, every tree path of F survives in $G - e$, so every vertex still has a path of length $T^*_G(v)$ to $v_{h*(v)}$. By (i) no time can fall, so $T^*_{G-e}(v) = T^*_G(v)$ for every v. The other two components of every label, the free beds and the identifier, do not depend on the roads, so every label, and with it every h^*, is unchanged. □

Proposition 18 (Greedy site selection is near-optimal). The coverage function f of Equation (26) is nondecreasing and submodular, with $f(\varnothing) = 0$. Therefore the set S_g chosen by Algorithm 4 satisfies

$$f(S_g) \geq \left[1 - \left(1 - \frac{1}{k}\right)^{k}\right] f(S^{opt}) \geq \left(1 - \frac{1}{e}\right) f(S^{opt}) \approx 0.632\, f(S^{opt}) \tag{32}$$

Source: Nemhauser et al. (1978), applied in this study to the coverage function of Definition 26

where S^{opt} is the best set of k sites.

Proof. $f(\varnothing) = 0$ because $U(\varnothing) = C_0$. Adding sites can only enlarge $U(S)$, so f never decreases. For $S \subseteq S'$ and a site $j \notin S'$, the gain from adding j to S is $\Sigma_{a \in Cj \setminus U(S)} P_a$, and since $U(S) \subseteq U(S')$, this is at least the gain $\Sigma_{a \in Cj \setminus U(S')} P_a$ from adding j to S'. That is submodularity. The bound in Equation (32) is the theorem of Nemhauser et al. (1978) for nondecreasing submodular functions under a limit on the number of elements chosen. □

Remark 12 (What the planning results buy). Proposition 17 cuts the work of a road-vulnerability study by more than half on the test graphs (Table 16), because a road segment off the forest cannot matter for that profile. Proposition 18 means that a provincial planner can choose Malasakit Center sites by a rule simple enough to do in a spreadsheet and still be sure of reaching at least 63.2 percent of the best possible coverage; in the tests the greedy rule did much better than its guarantee.

3.3 Software Verification

To keep programming errors out of the analysis, every proven property was also checked numerically against a separate reference implementation.

Row 2 of Table 14 deserves a close look.

Table 14. Transport pipeline checks, all produced by the project's test suite on random synthetic instances whose correct answers were obtained independently.

Property Checked	**Method of Check**	**Result**
Correctness of earliest arrival (Prop. 1)	618 random queries over 120 synthetic timetables, checked against complete enumeration of feasible journeys	577 reachable; arrival times agree in every case
Naive single-scan implementation	The same queries through a one-pass scan whose footpaths are not transitively closed	Late in 274 of 577 reachable queries (47.5%); mean error 32.5 min; worst 109 min
Gini scale invariance (Prop. 5 ii)	500 random weighted vectors, λ drawn from [0.2, 12]	Largest deviation 2.2×10^{-16}
Gini under translation (Prop. 5 iii)	500 random weighted vectors, c drawn from [0.5, 20]	Lower index in all 500 cases
Theil decomposition (Prop. 6)	500 random two-group partitions	Largest residual 5.6×10^{-16}
Vehicle-class monotonicity (Prop. 3)	300 random road graphs; restricted class made by deleting 35% of the arcs	Zero violations over 11,620 origin-destination pairs
Fare monotonicity and subadditivity (Prop. 7)	Distance sweep to 60 km and 1,000 random leg splits	Monotone over the sweep; subadditive in every split

Hospital mode and the agent network were checked the same way by `verify_hospital_mode.py`, run with a fixed seed.

Table 15. Checks on hospital mode and on the agent network (synthetic inputs only; no real hospital or mailbox).

Property Checked	**Method of Check**	**Result**
Early stopping, ambulance mode (Prop. 9, $k = 1$)	200 random road graphs (120 to 260 vertices, 6 to 15 hospitals, roughly 23% of arcs shut to the ambulance class); 4,000 random origin-profile queries against Floyd-Warshall distances and a set-based suitability filter	Reachable suitable hospital in 2,911 queries; hospital and travel time agree in all
Early stopping, patient mode (Prop. 9, $k = 1$)	120 random timetables (12 to 20 stops, 4 to 6 routes, 3 to 6 hospitals with sink vertices); 1,200 queries against a dynamic program in topological order	Reachable suitable hospital in 660 queries; full agreement; no arc runs backward in time
Nesting and monotonicity (Prop. 10 i to iii)	Each of 20 profiles at every ambulance query: 140,000 set-inclusion checks and 140,000 travel-time checks; 25,200 travel-time checks on the transit queries	No violation found
Payment gap (Cor. 11 a)	80,000 origin-profile pairs drawn from the ambulance instances	Never below zero; zero for every emergency profile

Property Checked	Method of Check	Result
Reader extraction (rule V2)	2,400 synthetic replies from 12 templates: 8 readable (reply form; English or Filipino free text with dates and telephone numbers as distractors) and 4 unreadable or contradictory	1,600 of 1,600 readable replies read exactly; 800 of 800 others referred to review; no wrong value
Message load (Prop. 12 i)	Mock mail environment, 60 hospitals, 30 days; 7,225 messages sent (4,902 inquiries and 2,323 reminders)	No hospital received more than 2 messages per cycle or 6 per day
Opt-out (Prop. 12 ii)	6 hospitals opting out after 2 to 10 messages	Nothing sent after an opt-out arrived
Report age (Prop. 12 iii)	24 hospitals replying within 2 h; age logged every 10 min	Oldest 579.8 min, inside the 600 min bound; never UNKNOWN after the first report
Validation (Table 11)	3,849 replies from the 51 responding hospitals, 67 of them with capacity typed ten times too large	3,302 accepted, none carrying a wrong value; 547 held, all 67 typing errors among them
Authentication (rule V1)	200 forged replies: never-issued tokens, genuine tokens from a look-alike address, or another hospital’s token	None accepted

Two rows of Table 15 need comment.

Table 16. Checks on the patient-needs and planning tools added in this revision, produced by `verify_patient_extensions.py` (seed 20261013; synthetic inputs only).

Property Checked	Method of Check	Result
Early stopping with backups, ambulance mode (Prop. 9, $k = 1$ and $k = 3$)	200 random road graphs (120 to 260 vertices; 6 to 15 hospitals with random levels, service lists, cost classes, and statuses; about 20% of road segments closed to the ambulance class); 4,000 random queries with 0 to 2 required services, against Floyd-Warshall distances and a set-based suitability filter	Suitable hospital reachable in 2,355 queries; the first hospital and the ranked list of three agree with the reference in all 4,000
Need monotonicity (Prop. 10 iv)	Every one-service enlargement of the need set of each query	28,037 set-inclusion checks and 28,037 travel-time checks; no violation
Early stopping, patient mode (Prop. 9)	120 random timetables (12 to 20 stops, 4 to 6 routes run both ways, 3 to 6 hospitals placed 0.1 to 0.8 km from a stop); 1,200 queries against the same search run to exhaustion	Agreement in all 1,200

Property Checked	Method of Check	Result
Slower walking (Prop. 13)	The same 1,200 queries for the standard class (4.5 km/h, 15 min cap) and the slow class (3.0 km/h, 10 min cap)	Slow walker never earlier. Where both reached a suitable hospital (673 queries), the slow walker was later in all 673 (mean +6.27 min, worst +68.4 min) and was sent to a different hospital in 151. In 280 more queries only the standard walker reached one
Catchment map from one search (Prop. 14)	60 random road graphs; Algorithm 2 against one early-stopping search from every vertex	Same hospital and travel time at all 11,474 vertices
Critical segments (Prop. 17)	Each of the 17,168 open road segments of the 60 graphs closed in turn and the map recomputed	8,576 segments lie on the forest, and closing any of them changed at least one vertex’s hospital or time; closing any of the other 8,592 changed nothing
Bed conservation (Prop. 15)	500 random instances (5 to 40 anchors, 2 to 10 hospitals, travel times 0 to 90 min, supplies 0 to 60 beds)	Largest residual 1.1×10^{-13}
Surge assignment (Prop. 16)	300 random instances on road graphs with 15 to 45 patients each (8,785 in all) and 0 to 6 free beds per hospital; Algorithm 3 against independent nearest choices and a first-come rule that serves emergencies first	Independent choices would have sent 2,722 patients to beds already taken. First-come placed 4,123 patients; the optimum placed 4,325. In the 170 instances where both placed the same numbers of emergency and other patients, the optimum was never slower and was faster in 121 (mean saving 17.16 min per instance). In the 13 instances where independent choices fit the beds, they equaled the optimum every time
Greedy site selection (Prop. 18)	496 random instances (30 anchors, 12 candidate sites, 2 existing class-0 hospitals, k = 1 to 4) against the best subset found by exhaustive search	Worst ratio 0.825, mean 0.9985; optimal in 96.6% of instances; guaranteed bound 0.632

Three rows of Table 16 say something about patients, not only about code.

3.4 Computational Cost

Table 17. Mean time per query for the verified engine; each configuration averages 40 queries.

Stops	Connections	Mean Query Time (ms)
50	900	1.03
100	2,288	1.41

Stops	Connections	Mean Query Time (ms)
200	5,620	3.66
400	12,576	15.06
800	28,336	69.51

Fitting log(time) against log $|C|$ gives a slope of 1.243 with $R^2 = 0.933$, in line with the near-linear growth of Proposition 2 once heap overhead is included.

Because its search ends early (Proposition 9), hospital mode was timed on its own.

Table 18. Early-stopping hospital search compared with a full search; random road graphs, 60 queries for each size, plain Python on one core.

Vertices	Arcs Open to the Ambulance	Hospitals	Early-Stopping Query (ms)	Full Search (ms)	Share of Vertices Settled
1,000	3,768	8	0.34	0.68	44.1%
4,000	14,898	26	0.61	3.58	13.8%
16,000	59,926	106	1.33	22.07	5.2%

On the largest graph, the full search settled about twenty times as many vertices as the early-stopping search and took about seventeen times as long.

The catchment map of Proposition 14 replaces one search per origin with a single search for the whole map.

Table 19. Building the catchment map with one search (Algorithm 2) compared with one early-stopping search per vertex; random road graphs with one hospital per 150 vertices, plain Python on one core.

Vertices	Road Segments	Hospitals	One Search for the Whole Map (ms)	Mean Per-Origin Search (ms)	Per-Origin Searches for All Vertices (ms)	Ratio
1,000	1,863	6	0.83	0.055	55	65.9
4,000	7,473	26	5.86	0.272	1,087	185.6
16,000	29,821	106	39.08	0.206	3,300	84.4

The times in Table 19 were measured on a different machine from those in Tables 17 and 18, so the two sets should not be compared directly.

3.5 Worked Instance S1

Network S1 is a small synthetic test bed built so that every stage of the pipeline can be traced by hand, with ground-truth answers checked independently.

Table 20. Instance S1 in brief.

Element	**Value**
Stops	7 (S1 to S7)
Routes	4 base routes, run in both directions
Connections in the service day	787
Minimum change time θ_{ch}	4 min
Walking cap $\bar{t}_W$	15 min
Footpaths	S3 to S6 (11 min), S2 to S5 (13 min)
Departure grid $\mathcal{T}_0$	06:00 to 08:00 every 5 min (25 departure times)
Baseline budget θ	30 min
Destination layer	hospital beds at S4 (120), S7 (40), and S3 (25)
Anchors	six, A1 to A6, split between municipalities M1 and M2

Table 21. Door-to-door travel time (min), averaged over the departure grid, from each anchor of S1 to each hospital site, with the resulting 30-minute accessibility in reachable beds.

Anchor	**Weight**	**Municipality**	**to S4 (120)**	**to S7 (40)**	**to S3 (25)**	A_a**(30)**
A1	4,200	M1	29.2	49.0	24.6	145
A2	3,100	M1	27.3	61.0	21.3	145
A3	1,850	M1	18.6	51.4	7.0	145
A4	1,400	M2	43.6	76.6	37.5	0
A5	980	M2	32.6	65.8	26.0	25
A6	2,600	M2	37.2	13.0	48.1	40

Table 22. Inequality of access to hospital beds in S1 at the 30-minute baseline (synthetic instance).

Quantity	**Value**
Weighted mean μ	102.99 beds

Quantity	Value
Gini coefficient G	0.2747
Theil index T	0.2205
Part within municipalities	0.0308 (14.0% of total)
Part between municipalities	0.1897 (86.0% of total)
Identity check $\lvert T - (\text{within} + \text{between}) \rvert$	2.8×10^{-17}
Palma ratio	0.907
Weight share with zero access	9.9%

Anchor A1, 29.2 min from the 120-bed hospital, falls just inside the budget, while A5, at 32.6 min, falls just outside it.

The decomposition works as intended.

Table 23. Leg fares under the traditional jeepney parameters of Table 6.

Leg Length d (km)	Full Fare (₱)	Fare with 20% Discount (₱)
1.0	14.00	11.20
4.0	14.00	11.20
4.2	16.00	12.80
7.0	20.00	16.00
10.5	28.00	22.40
18.0	42.00	33.60

Table 24. Independent tests on new random instances and on S1.

Property	Method	Result
Correctness of earliest arrival (Prop. 1)	400 random queries on S1 against exhaustive enumeration	400 reachable; full agreement
Labeling by stop only (Remark 1)	The same 400 queries with one label kept per stop	Wrong in 6 of 400 (1.5%); mean error 15.0 min; worst 30 min
Gini scale invariance (Prop. 5 ii)	500 random weighted vectors, λ in [0.2, 12]	Largest deviation 1.1×10^{-16}
Gini under translation (Prop. 5 iii)	500 random weighted vectors, c in [0.5, 20]	Lower in all 500
Theil decomposition (Prop. 6)	500 random three-group partitions	Largest residual 1.7×10^{-16}

Property	Method	Result
Vehicle-class monotonicity (Prop. 3)	60 random road graphs, width threshold raised from 1.3 m to 2.2 m	Zero violations in 5,400 origin-destination pairs
Fare monotonicity and subadditivity (Prop. 7)	Sweep from 0 to 60 km in 0.1 km steps and 1,000 random leg splits	Monotone throughout; subadditive in every split

Row 2 shows the second implementation pitfall found in this study, and it is much harder to spot than the first.

3.5.1 Hospital Mode on Instance S1

To run hospital mode from end to end, Table 25 assigns invented attributes to the three hospital sites of S1.

Table 25. Synthetic hospital attributes created for S1.

Site	c_h (beds)	η_h	Ownership and Programs	q_h	$\hat{o}_h$	b_h	σ_h
S4	120	3	Government; with Malasakit Center	0	112	8	AVAILABLE
S7	40	2	Private; accredited by PhilHealth	2	31	9	AVAILABLE
S3	25	1	Government; without Malasakit Center	1	25	0	FULL

Table 26. Recommendation and travel time (min) for each anchor of S1 under several patient profiles ($N = \varnothing$, standard walker).

Anchor	Plain Nearest	κ = 1, any ρ	κ = 2, ρ = 0	κ = 3, ρ = 0	κ = 3, ρ = 3	κ = 5, ρ = 1
A1	S3, 24.6 (full)	S4, 29.2	S4, 29.2	S4, 29.2	S4, 29.2	S4, 29.2
A2	S3, 21.3 (full)	S4, 27.3	S4, 27.3	S4, 27.3	S4, 27.3	S4, 27.3
A3	S3, 7.0 (full)	S4, 18.6	S4, 18.6	S4, 18.6	S4, 18.6	S4, 18.6
A4	S3, 37.5 (full)	S4, 43.6	S4, 43.6	S4, 43.6	S4, 43.6	S4, 43.6
A5	S3, 26.0 (full)	S4, 32.6	S4, 32.6	S4, 32.6	S4, 32.6	S4, 32.6
A6	S7, 13.0	S7, 13.0 (stabilize), then S4, 37.2	S7, 13.0	S4, 37.2	S7, 13.0	S4, 37.2

Table 26 shows four effects.

The second effect is capability.

The third effect is cost.

The fourth effect comes from the two emergency rules.

Table 27. Beds reachable within 30 min in S1, counted three ways: authorized, free, and free in hospitals that payment class 0 accepts (synthetic instance).

Quantity	**Authorized, $A_a(30)$**	**Free, $B_a(30)$**	**Free and Accepted, $B^0_a(30)$**
Scores, A1 to A6	145, 145, 145, 0, 25, 40	8, 8, 8, 0, 0, 9	8, 8, 8, 0, 0, 0
Weighted mean μ	102.99	6.84	5.18
Gini coefficient	0.2747	0.1859	0.3524
Theil index	0.2205	0.1857	0.4345
Share within municipalities	14.0%	84.8%	0.0%
Share between municipalities	86.0%	15.2%	100.0%
Palma ratio	0.907	0.486	2.103
Weight share with zero access	9.9%	16.8%	35.2%

Table 27 shows how the diagnosis shifts when free beds replace authorized ones.

3.5.2 Patient Needs and Backups on S1

Table 28 adds the patient's needs and the backup hospital to the same instance.

Table 28. First choice and backup (min) for each anchor of S1 by the patient's need and payment class, with κ = 3; "none" means no second suitable hospital exists.

Anchor	**No particular need, ρ = 3**	**Maternity or high-risk pregnancy (MAT or HRP), ρ = 3**	**Dialysis (DIA), ρ = 3**	**Any need, ρ = 0**
A1	S4, 29.2; backup S7, 49.0	S4, 29.2; backup S7, 49.0	S4, 29.2; none	S4, 29.2; none
A2	S4, 27.3; backup S7, 61.0	S4, 27.3; backup S7, 61.0	S4, 27.3; none	S4, 27.3; none
A3	S4, 18.6; backup S7, 51.4	S4, 18.6; backup S7, 51.4	S4, 18.6; none	S4, 18.6; none
A4	S4, 43.6; backup S7, 76.6	S4, 43.6; backup S7, 76.6	S4, 43.6; none	S4, 43.6; none
A5	S4, 32.6; backup S7, 65.8	S4, 32.6; backup S7, 65.8	S4, 32.6; none	S4, 32.6; none

Anchor	No particular need, ρ = 3	Maternity or high-risk pregnancy (MAT or HRP), ρ = 3	Dialysis (DIA), ρ = 3	Any need, ρ = 0
A6	S7, 13.0; backup S4, 37.2	S7, 13.0; backup S4, 37.2	S4, 37.2; none	S4, 37.2; none

The maternity column matches the first column only because S3 is full.

The backup column carries a warning that a map of hospitals alone would never give.

3.5.3 Beds per Person on S1

Table 29 applies the bed-to-population ratio of Definition 23 to S1, first with authorized beds and then with free beds.

Table 29. Beds per 10,000 people in S1 by the two-step floating catchment area method (Definition 23) at θ = 30 min, with authorized and with free beds (synthetic instance).

Quantity	Authorized Beds	Free Beds
R_h for S4, S7, S3	131.15, 153.85, 24.68	8.74, 34.62, 0.00
A^F_a for A1 to A6	155.83, 155.83, 155.83, 0.00, 24.68, 153.85	8.74, 8.74, 8.74, 0.00, 0.00, 34.62
Weighted mean	130.93	12.03
Beds counted, $\Sigma_a P_a A^F_a$ (Prop. 15)	185	17
Gini coefficient	0.1585	0.4247
Theil index	0.1472	0.4093
Share within municipalities	75.8%	84.1%
Weight share with zero access	9.9%	16.8%

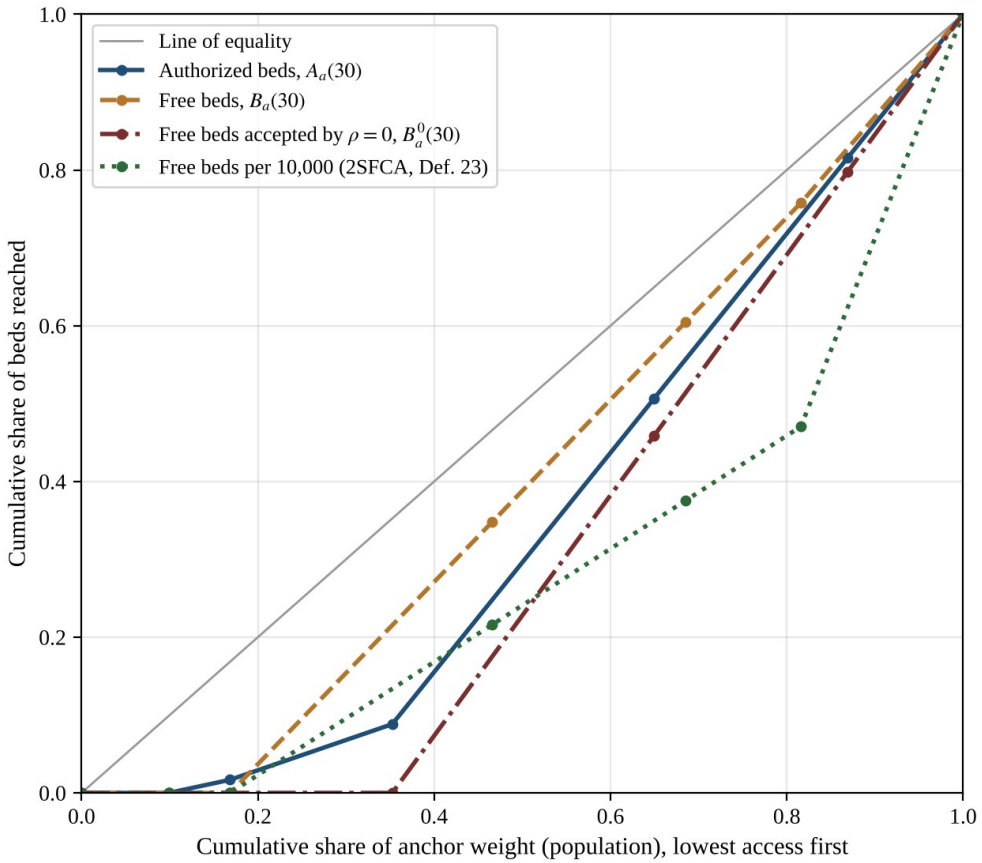


Figure 4. Lorenz curves of bed access in S1 at the 30-minute baseline, counted four ways. The farther a curve sags below the line of equality, the more unequal the access. Synthetic instance.

The two ways of counting free beds tell different stories.

With authorized beds the movement goes the other way: the Gini coefficient falls from 0.2747 to 0.1585, because A1 to A3 reach many beds but share them among many people.

3.5.4 Surge Assignment on S1

Suppose a road crash leaves 15 emergency patients (κ = 2, so cost does not filter) spread over the six anchors, as in Table 30.

Table 30. Surge assignment on S1: 15 emergency patients (κ = 2) and 17 free beds.

Anchor	Patients	Independent Choice	Optimal Assignment (Algorithm 3)	Travel Time Each (min)
A1	4	S4	S7	49.0 (S4 would be 29.2)
A2	3	S4	S4	27.3
A3	2	S4	S4	18.6

Anchor	Patients	Independent Choice	Optimal Assignment (Algorithm 3)	Travel Time Each (min)
A4	2	S4	S4	43.6
A5	1	S4	S4	32.6
A6	3	S7	S7	13.0
Total	15	S4: 12 patients for 8 beds; S7: 3	S4: 8; S7: 7	473.9 in all

If every patient follows his or her own nearest-hospital advice, 12 patients head for the 8 free beds of S4 and 4 of them are turned away at the door.

A first-come rule would leave the outcome to chance: whichever patients called last would be sent to S7, and if those were the patients from A4, each would travel 76.6 minutes instead of 43.6.

3.6 Computational Experiments on the Open-Data Instance

These experiments run on the data set that the pipeline of Section 3.1 builds for Albay, together with randomized variants of it.

3.6.1 Sensitivity Analysis

The sensitivity tests vary the anchor set, the headways, the vehicle-class limits, the departure-time assumptions, and the travel-time budget.

Hospital mode is re-run under two alternative acuity tables (one lowering each $\lambda(\kappa)$ by one level, the other requiring Level 3 care for both emergency levels), under several stabilization margins and freshness limits, under both mobility classes, and with service sets taken from level defaults alone or adjusted by register entries (Equation 12).

3.6.2 Statistical Treatment

Eleven questions are tested formally: six on the transport layer and five on the hospital layer.

Table 31. How each question is tested.

Question	Test or Estimator	Reason for the Choice
(S1) Is modeled access more unequal than resampling noise alone would make it?	Weighted Gini (Gini, 1912) with a 95% percentile bootstrap interval from 2,000 resamples (Efron, 1979; Efron & Tibshirani, 1993)	Assumes no distribution; the interval reflects the sampling noise that comes from the anchor set
(S2) Do urban and rural anchors differ?	Mann-Whitney U (Mann & Whitney, 1947), with the rank-biserial correlation as effect size (Kerby, 2014)	Access is skewed to the right and floored at zero, which rules out a t-test
(S3) Is the inequality local or structural?	Theil decomposition (Prop. 6; Shorrocks, 1980) with a bootstrap interval for the within-group share	An exact decomposition separates the two explanations cleanly
(S4) Does modeled access track settlement density?	Spearman rank correlation (Spearman, 1904)	No linear relation is assumed
(S5) How sensitive is the result to the anchor set?	Median discrete Fréchet distance between Lorenz curves (Eiter & Mannila, 1994)	Compares whole curves rather than single points
(S6) How sensitive is the result to the headway assumption?	Spread of results when headways are perturbed by ±25% and ±50%	Headway is the least constrained input of the model
(S7) Do urban and rural anchors differ in payment gap?	Mann-Whitney U with the rank-biserial effect size (Kerby, 2014)	The gap is nonnegative and has many zeros
(S8) How often is the plain nearest hospital unsuitable?	Weighted share of anchors with a 95% percentile bootstrap interval from 2,000 resamples (Efron & Tibshirani, 1993)	A proportion is better described by an interval than by a test
(S9) How large is the need gap for dialysis?	Weighted share of anchors with $\Delta_{need} > 15$ min, with a bootstrap interval	Same reason as S8
(S10) Are a few roads responsible for most of the risk?	Share of open road segments that carry half of the total finite criticality $\Sigma_e D_e$, with segments of infinite D_e counted separately	Concentration is read directly from the sorted criticality values
(S11) How much access do slow walkers lose?	Paired bootstrap interval for the difference in weighted share of anchors with no suitable hospital within 60 min, slow minus standard class	The same anchors are compared under both classes, so the pairs are kept

3.6.3 Pre-Registered Model Hypotheses

All hypotheses and decision rules below were fixed before the study-area instance was run.

- **M1.** Modeled access to hospital beds has a weighted Gini above 0.30 at the 30-minute baseline. The hypothesis holds if the bootstrap interval lies wholly above 0.30.
- **M2.** Urban-center anchors have greater modeled access than rural ones, tested with the Mann-Whitney U at $\alpha = 0.05$.
- **M3.** The between-municipality term is under half of the total, which would make modeled transport disadvantage chiefly a matter of local feeder service.
- **M4.** Larger vehicle classes never use more of the road network. Proposition 3 makes this exact rather than typical, so a single violation would point to a faulty road attribute and send that segment back for checking.
- **M5.** Under every anchor set of Section 3.6.1 and every $\theta \in \{15, 30, 45\}$ min, hypotheses M1 to M3, M6, and M7 point the same way.
- **M6.** With $\kappa = 3$ and $\rho = 0$ at the 30-minute baseline, the payment gap is larger at rural anchors than at urban ones, tested with the Mann-Whitney U at $\alpha = 0.05$.
- **M7.** With $\kappa = 2$, more than half of the anchor weight has a plain nearest hospital that fails the capability test of Definition 17. The hypothesis is supported if the bootstrap interval lies wholly above 0.5.
- **M8.** For $\kappa = 1$ and $\kappa = 2$, no anchor has a nonzero payment gap. Corollary 11 makes this exact, so any nonzero value would reveal a coding error.
- **M9.** With $\kappa = 3$, $\rho = 3$, and $N = \{\text{DIA}\}$, more than half of the anchor weight has a need gap above 15 minutes. The hypothesis is supported if the bootstrap interval lies wholly above 0.5.
- **M10.** For $\kappa = 2$ in ambulance mode, fewer than 5 percent of the open road segments carry half of the total finite criticality.
- **M11.** The weighted share of anchors with no suitable hospital ($\kappa = 3$, $\rho = 3$) within 60 minutes by public transport is at least 10 percentage points higher for the slow class than for the standard class; supported if the paired bootstrap interval lies wholly above 0.10.

Within the model, M2 and M6 measure urban-rural gaps relevant to targets 10.2 and 11.2.

3.6.4 Sensitivity Sweeps

Seven sweeps recompute the main results across parameter ranges: the travel-time budget; departure windows in the morning peak, at midday, and in the evening; the decay parameter of Definition 8, set to half-lives of 15, 30, or 45 minutes; a half-headway wait model; each of the three anchor sets; the mobility classes; and, for hospital mode, the alternative acuity tables, freshness limits, stabilization margins, and service-set rules of Section 3.6.1.

4. CONCLUSIONS

4.1 Evaluation of the Findings

The primary question asked how quickly a patient starting from any anchor in Albay can reach a hospital that can treat the patient, has a free bed, and accepts what the patient can pay, and how evenly that access is shared. The study's answer is that this quantity can be defined precisely, computed exactly, recomputed in milliseconds when a bed report or a road changes, and studied before any field survey.

Objective 1: the hospital map and the transport graphs. The label-setting search on the time-expanded graph returns exact earliest arrivals (Proposition 1), and it matched brute-force enumeration in all 1,018 test queries (Tables 14 and 24). The vehicle classes form a partial order, so a larger vehicle never has more access than a smaller one (Proposition 3, Corollary 4), and the van-type ambulance sits between the sport utility vehicle and the jeepney in every limit (Remark 10).

Objective 2: the patient profile and the suitability rule. A patient's acuity, required services, payment class, and mobility combine into one rule (Definition 17) whose behavior is provably orderly: more money never removes a hospital, more needs never add one, and an emergency patient is never filtered by ability to pay (Proposition 10). The tests found no violation in 140,000 set-inclusion checks for payment and acuity, 28,037 for needs, and 80,000 payment-gap checks (Tables 15 and 16).

Objective 3: the nearest suitable hospital and its backups. Early stopping loses no exactness, for one hospital or for a ranked list (Proposition 9). Hospital routing matched the independent

references in all 5,200 queries of the original tests and all 5,200 new ones, and on a 16,000-vertex graph the search settled only 5.2 percent of the vertices.

Objective 4: the agent network. All three bounds of Proposition 12 held over 30 simulated days: no hospital received more than 2 messages in a cycle or 6 in a day, nothing was sent after an opt-out, and every prompt responder’s report stayed younger than 10 hours. All 200 forged replies were rejected, and all 67 typing errors were held for review.

Objective 5: catchments and bed access. The whole catchment map comes from one search (Proposition 14). It agreed with per-vertex searches at all 11,474 test vertices and took about 39 ms for a 16,000-vertex network.

Objective 6: surge assignment. The assignment program has whole-number optimal solutions that never overfill a hospital and that place the most patients possible, emergencies first (Proposition 16). On S1 it placed all 15 crash victims where independent choices would have turned 4 away.

Objective 7: critical roads and site selection. Only road segments on the shortest-path forest can delay patients when they are closed (Proposition 17), which halved the segments that had to be tested. Greedy site selection is guaranteed at least 63.2 percent of the best coverage (Proposition 18); in the tests it never fell below 82.5 percent and was exactly optimal in 96.6 percent of instances.

Objective 8: verification. Two natural programming shortcuts were tested against independently derived answers, and both failed. A single scan without transitively closed footpaths arrived late in 47.5 percent of reachable queries, by 32.5 minutes on average.

The synthetic example makes the patient-level stakes concrete. In network S1, routing by distance alone sent patients from five of the six anchors to a hospital with no free bed.

4.2 Limitations

- **(L1) Headways stand in for schedules.** Many informal operators leave only when the vehicle is full, so departures are irregular. Modeling them as a regular series at an estimated headway hides that, and headway is the least supported input of the model. The

half-headway wait model and the ±50 percent perturbations of Table 31 bound the effect but do not remove it.

- **(L2) Inferred routes versus driver behavior.** Routes on the road network are fixed by vehicle-class limits and by terminal locations. Drivers may take shortcuts, change roads in quiet hours, or change routes on market days, and the model does not capture this. It is the largest difference between the model and real conditions, and it is why field validation is essential rather than optional (Section 4.3).
- **(L3) Incomplete registers.** Coverage is much better in town centers than in remote areas. The health station of an upland barangay may be missing from every national register, and even if it is listed, OpenStreetMap is unlikely to hold the small paths up the hill or the limits of the bridges along them. The resulting bias makes access in hard-to-reach places look better than it is, and the model cannot correct it.
- **(L4) No network outages.** A typhoon, a Mayon Volcano alert, or a local flood can close roads for days, but the feed models only normal weekday and Saturday service. Proposition 17 identifies which closures matter; the feed itself does not simulate them.
- **(L5) Imperfect destination data.** Facility records lag behind actual conditions. Bed and classroom counts are reported by the facilities themselves without independent audit, and new schools or health units may be missing.
- **(L6) Choice of anchors.** Landmark anchoring replaces the modifiable areal unit problem with a selection problem: adding or dropping one terminal changes the index set and the inequality statistics built on it. The inclusion criteria of Appendix C were fixed before listing began, the anchor list is published with the feed, and Section 3.6.1 repeats the analysis on three anchor sets.
- **(L7) Provincial border.** Journeys that cross into Sorsogon or Camarines Sur, or reach Catanduanes by ferry, are cut at the border, which understates access for anchors near it. Hospital mode shares this limit and will never suggest a suitable hospital just across the border.
- **(L8) Synthetic nature of S1.** Tables 20 to 30 show that the pipeline follows its own rules; they say nothing empirical about Albay. In particular, the Gini coefficient of

0.2747 in Table 22 comes from a seven-stop synthetic network and must not be read as a first estimate for the province.

- **(L9) Voluntary hospital reporting.** The network can ask but cannot confirm reported counts or demand a reply. A wrong count that passes the plausibility checks stays in force until a later report replaces it, and hospitals that never answer stay UNKNOWN, a safe but uninformative state.
- **(L10) Acuity and needs are entered, not assessed.** The application does not examine patients and is not a clinical device. Acuity levels and required services are chosen by ambulance crews or users. Clinical judgment always overrides the application, and the parameters K_E, λ, and θ_{stab} are placeholders until the Provincial Health Office sets them.
- **(L11) Ordinal cost ranking.** Equation (11) ranks hospitals by expected out-of-pocket burden using only ownership and program flags. It says nothing about actual bills, the size of Malasakit Center assistance, or specific PhilHealth benefits.
- **(L12) Free-flow ambulance times.** Ambulance times assume unobstructed traffic (Assumption 8). They leave out traffic jams, emergency driving, dispatch queues, and handover delays.
- **(L13) Simulated e-mail.** Real e-mail meets bounces, spam filters, shift changes, and firewalls that a mock server does not. The extraction rules were tested on templates written together with the rules, so their success rate on real replies is unknown. Unreadable messages always go to human review.
- **(L14) Synthetic hospitals in S1.** The hospital figures in Tables 25 to 30 were made up to demonstrate hospital mode and contain no information about real facilities.
- **(L15) Service lists.** Service sets come from level defaults and from published register entries (Equation 12). A hospital may lack a listed service on a given day, for example when a dialysis machine is under repair, or may offer a service that no register lists. The agents ask only about beds and the emergency room, not about services.
- **(L16) Surge assignment.** The assignment assumes that free beds are known and that requests arrive together (Assumption 10). Real requests arrive over time, the number of ambulances is limited, and patients within the same acuity group may differ in urgency. The assignment is advice for a dispatcher, not an order.

- **(L17) Two mobility classes.** Walking ability varies far more than two classes allow. The model has no data on stairs, steep paths, or wheelchair access at stops, and the slow-class values are declared, not measured.
- **(L18) Coverage-based site selection.** Definition 26 counts people who come within reach of an assisted-care hospital. It ignores the cost of setting up a Malasakit Center, the number of beds at the chosen hospital, and the crowding that new users would bring. Proposition 15's ratios should be checked for any site the greedy rule proposes.

4.3 Recommendations and Challenges in Extending the Results

Regional extension. Scaling the framework beyond one province is an operational challenge, not a computational one. Proposition 2 and the times in Tables 17 to 19 show that an ordinary laptop can handle much larger feeds and road networks.

Field validation. The obvious next step is to test the model in the field. The workflow was designed so that field data can enter without disturbing the rest of the pipeline: a ground-surveyed feed can replace the generated feed of Stage 3 while the routing logic of Section 3.2 and all earlier stages stay the same.

Operating hospital mode. Moving hospital mode from simulation to live use raises consent and governance requirements. Three conditions must be met before the agents write to any hospital: the Provincial Health Office must be formally informed, the Scientific Review Committee must approve the protocol, including the message of Appendix F and the data rules, and each facility must opt in through its public contact channel.

Patient needs and planning tools. The researchers recommend four further steps. First, the service lists of Albay's hospitals should be checked with the Provincial Health Office, and a question on services available today should be added to the inquiry only after the Scientific Review Committee approves the change.

4.4 Applications

For patients and families. A family can see, before leaving home, which hospital can treat the patient, whether it has a free bed and how old that report is, what it costs, how to get there by jeepney or tricycle, what the fare will be with the discount for a senior citizen, student, or person

with disability, and where to go next if the first hospital turns them away. A family that can use only free or assisted care sees the nearest hospital it can actually use, not the nearest hospital overall.

For ambulance crews and barangay health workers. Crews can choose a hospital and justify the choice before the vehicle moves, avoid roads closed to their van, see backup hospitals with the extra minutes each costs, follow the stabilization prompt for the most critical patients, and call 911 with one tap.

For the Provincial Health Office and the PDRRMO. During typhoons and Mayon Volcano alerts, the health office can watch free beds by municipality, redraw hospital catchments within milliseconds whenever a status changes, and use the surge assignment to spread many patients across the free beds instead of letting ambulances converge on one hospital. Before a typhoon, the PDRRMO can use the critical-road list to decide which segments to clear first and where to pre-position equipment.

For LGU and DOH planners. Bed deserts, payment-gap maps, and bed-to-population ratios show which communities are far from affordable and suitable care. Greedy site selection proposes where a new Malasakit Center would bring the most low-income people within reach, and recomputing the ratios of Definition 23 shows where added beds would relieve the most crowding.

Alignment with the Sustainable Development Goals. Seen through the 2030 Agenda, each application relates to a particular target, even though real-world use is still some way off. Hospital mode and the bed table bear on targets 3.8 and 3.d, and needs-aware routing for maternity, high-risk pregnancy, and newborn care on targets 3.1 and 3.2.

Scalability. The framework is not tied to Albay. It uses only OpenStreetMap, public facility registers, open population data, and ordinary hardware, so any province that depends on informal transport could apply it.

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APPENDIX A. OPEN DATA SOURCE REGISTER

Every input of the model traces to one row of Table A1.

Table A1. Where each layer of the model comes from.

Source	Publisher	Layer Filled	Access
OpenStreetMap extract for the Bicol Region (OpenStreetMap contributors, 2026)	OpenStreetMap Foundation	Road and walking graphs, road attributes, junction anchors	Open Database License
2020 Census of Population and Housing (PSA, 2021)	Philippine Statistics Authority	Barangay populations behind the anchor weights	Public release
Philippine Standard Geographic Code (PSA, n.d.)	Philippine Statistics Authority	LGU and barangay codes; barangay count	Public web service
Land areas of LGUs (PhilAtlas, n.d.)	PhilAtlas, from PSA figures	Land area in Table 3	Public web page
National Health Facility Registry (DOH, n.d.-b)	Department of Health	Hospitals, infirmaries, and rural health units, with ownership and authorized beds	Public directory
Hospital licensing records under the DOH (2012) classification	Health Facilities and Services Regulatory Bureau, Department of Health	Service level η_h, emergency-room flag e_h, and register entries X^+_h, X^-_h for the service set (Equation 12)	Public listing
List of accredited hospitals and infirmaries (Philippine Health Insurance Corporation, 2026)	Philippine Health Insurance Corporation	Accreditation flag for the cost class q_h	Public listing
Malasakit Center directory (DOH, n.d.-a)	Department of Health	Malasakit flag for the cost class q_h	Public listing
Port listings (Philippine Ports Authority, n.d.)	Philippine Ports Authority	Ports and piers used as anchors	Public listing
Contact addresses of facilities	Registry entries and official facility websites	Address m_h for the inquiry in Appendix F	Public, per facility
Masterlist of Schools (Department of Education, 2021)	Department of Education	Public elementary and secondary schools, with classroom counts where available	Public release
Fare adjustment effective September 28, 2026 (Daguno-Bersamina, 2026; Nieto, 2026)	Land Transportation Franchising and Regulatory Board	Fare parameters of Table 6	Public announcement

Source	Publisher	Layer Filled	Access
Route-plan annexes of LGUs, where published (DILG, 2018; DOTr, 2017)	Municipal and city planning offices	Terminal anchors and named corridors	Depends on the LGU
Disaster risk reduction plans of LGUs, where published	Municipal and city DRRM offices	Designated evacuation centers	Depends on the LGU
Bed-status replies (live operation only)	Each participating hospital	Reported capacity, occupancy, and emergency-room status (Definition 15)	Only with the facility's consent; none used here

APPENDIX B. GTFS FILES PRODUCED

Table B1. Contents of the feed and of its extension files.

File	Contents
agency.txt	Operator or association name, service area, contact details, time zone
routes.txt	Route ID, short and long names, mode
trips.txt	A record for each modeled vehicle run
stops.txt	Boarding points inferred by the model, with names and coordinates
stop_times.txt	Arrival and departure time at each stop of each trip
calendar.txt	Service days, with weekday and Saturday patterns kept apart
shapes.txt	Generated route geometry
frequencies.txt	Headway-defined service wherever departure times are not fixed
fare_attributes.txt	Per mode: base fare, base distance, and rate per succeeding kilometer
fare_rules.txt	How fares attach to routes and zones
landmarks.txt (extension)	Anchor and destination inventory, with class, source register, and size w_p
hospitals.txt (extension)	Hospital records of Definition 13: service level, service set, emergency-room flag, ownership, PhilHealth and Malasakit flags, cost class, and contact address
bed_status.txt (extension, live operation only)	Latest validated report of each hospital: time stamp, reported capacity and occupancy, emergency-room status, free beds, and status. It is served by the Status agent and is never part of the published feed.

APPENDIX C. RULES FOR INCLUDING LANDMARKS AND THE INVENTORY SCHEMA

The criteria below were settled before the first landmark was listed.

Inclusion criteria. To enter the landmark set, a place must meet all four conditions.

1. At least one register in Table 4 lists it, under a name familiar to residents.
2. Its published coordinate lies no more than 150 m from a road-network node, so that its entrance node can be assigned mechanically.
3. It is operating now. Facilities that their registers mark as closed, proposed, or under construction are left out.
4. Destinations need a size. The register supplies w_p, or else $w_p = 1$ and the destination is simply counted; one category never mixes counted and sized destinations.

Exclusion and merging rules. Private schools go into a separate layer, since the atlas concerns the capacity the government provides.

Table C1. Fields of the landmark inventory, with the extra fields kept for hospitals.

Field	**Type**	**Description**
landmark_id	string	Permanent identifier given by the project
name	string	Name as written in the source register
class	enum	A class from Table 4
role	enum	anchor, destination, or both
lat, lon	decimal	Coordinate as published (WGS 84)
entrance_node	integer	Road node serving as o_a or as the destination point
snap_distance_m	decimal	Meters between the published coordinate and the entrance node
size_w	integer	Beds, classrooms, or service windows; 1 for a counted destination
lgu_code	string	Philippine Standard Geographic Code of the LGU that contains the landmark
source	string	Register that the record came from
source_date	date	Date on which the register was consulted
merge_log	string	Identifiers of records merged into this one
Hospitals and infirmaries only		
service_level	integer	η_h: 0 for an infirmary, 1 to 3 for a hospital level (Table 7)
services	list	Σ_h: codes from Table 8, after Equation (12)
services_source	string	Register entry behind any change from the level default, with its date

Field	Type	Description
er_flag	boolean	e_h: whether an emergency room is present
ownership	enum	government or private
philhealth_accredited	boolean	Whether the facility is on the PhilHealth accreditation list
malasakit_center	boolean	Whether a Malasakit Center is present
cost_class	integer	q_h by Equation (11)
contact_email	string	Official e-mail address of the facility (never a personal one)

APPENDIX D. APPLICATION FEATURE CHECKLIST

Each feature ends with the SDG target or targets it is meant to support (UNGA, 2015).

Hospital mode

1. A hospital-mode switch that leaves only hospitals and infirmaries on the map and changes the screen to step-by-step guidance toward a hospital (SDG 3.8).
2. Two travel modes: the ambulance, which uses the ambulance class of Table 5 on roads, and the patient, who rides public transport and walks.
3. Plain-language choices for the five acuity levels of Table 12, made by the crew or by the user.
4. A needs selector that lists the eight services of Table 8 in plain words, with “no particular need” as the default (SDG 3.1, 3.2, and 3.8).
5. A mobility switch (“I walk slowly”) that applies the slow class of Definition 16 to every walking leg (SDG 11.2 and 10.2).
6. Payment-class filtering among the four classes of Table 9. The filter is switched off automatically at acuity levels 1 and 2, so emergency routing is never delayed by cost (SDG 3.8 and 1.4).
7. A hospital information card showing the route, travel time, care level, services, cost class, free beds, and the age of the report. An UNKNOWN status is shown as unknown, never as available (SDG 3.8 and 3.d).
8. A backup list showing the next two suitable hospitals and the extra minutes each would cost (Definition 18).

9. A stabilization prompt: for acuity level 1, if an open emergency room is more than θ_{stab} nearer than the nearest suitable hospital, the application advises stabilization there first (Definition 19; SDG 3.8).
10. A one-tap shortcut to call 911 on every hospital-mode screen, in line with Executive Order No. 56 (2018).
11. A read-only dashboard for the Provincial Health Office showing free beds and report age by municipality (SDG 3.d).
12. Catchment and bed-desert layers on the dashboard, drawn for a chosen patient profile (Definitions 22 and 23; SDG 3.8 and 10.2).
13. A surge view on the dashboard that takes a list of waiting patients and proposes an assignment by Algorithm 3 for a dispatcher to confirm (Definition 24; SDG 3.d and 11.5).
14. A critical-road list for the PDRRMO, ranked by Definition 25, for use before and after typhoons and eruptions (SDG 11.5 and 13.1).

Transport features

15. Door-to-door journey planning inside the study area, telling the user when they will arrive, how much they will pay, which vehicle to take, and where to get on and off (SDG 11.2).
16. A discount switch for senior citizens, students, and persons with disability (SDG 11.2 and 10.2).
17. Walking directions for the first and last mile, with a cap on walking time (SDG 11.2).
18. A choice among the ten vehicle classes of Table 5, giving a road route that the class can use and marking any restricted segments (SDG 9.1 and 11.5).
19. A fare lookup between any two named landmarks (SDG 11.2).
20. A 30-minute landmark accessibility map, with a slider for other budgets (SDG 11.2, 10.3, and 4.a).
21. Offline storage of the road extract and the feed for times of weak signal, such as typhoons (SDG 11.5 and 13.1).

22. A button for reporting errors, so that riders can help fix the dataset once it is released. This is the only route by which field observations could reach the transport dataset, and only after publication; corrections are logged by date, and none was used in this paper (SDG 17.18).

APPENDIX E. RISK REGISTER

The work poses no physical risk to human subjects.

Table E1. Risks and how each is handled.

Risk	Likelihood	How It Is Handled
Model output mistaken for real-world measurement	High	Every computed value is labeled a model output. No synthetic result is offered as a finding about a real facility, and the provenance note of Section 3.0 is kept.
Generated routes differ from actual operator routes	High	Recognized as Limitation L2. Routes are anchored to published terminals and markets; comparison with surveyed feeds is left to field validation.
Missing road attributes in OpenStreetMap	High	Missing attributes are set to +∞ and flagged. The share of flagged segments is reported beside every vehicle-class result.
Hospital does not respond, so its status stays unknown	High	Non-responding hospitals stay in the suitable set with status UNKNOWN. The age of the data is shown, and unknown capacity is never shown as available (Definition 17).
Landmark register is out of date	Moderate	Source and access date are recorded for each entry (Appendix C). Coverage is reported by municipality, and gaps are read as understating disadvantage (Limitation L3).
Unchecked software shortcut produces wrong routes	Moderate	All routing outputs were checked against independently written reference code before analysis (Tables 14, 15, 16, and 24).
Inequality measures sensitive to the choice of anchors	Moderate	Inclusion criteria were fixed before listing (Appendix C). Three anchor sets are compared (Section 3.6.1), and full anchor lists are released with the data.
Typing or parsing errors in bed counts	Moderate	Rules V2 to V4 hold unclear replies for human review. All 67 simulated typing errors were caught in testing (Table 15).
Hospital staff see the inquiries as a nuisance	Moderate	At most two messages per cycle (Proposition 12), a once-a-day option, and immediate handling of opt-outs before the next send.
Service list out of date, so a patient is sent to a hospital that lacks the service today	Moderate	Each change from the level default is tied to a dated register entry (Appendix C). The card shows backups and reminds the crew to call ahead; the 911 button is always present (Limitation L15).

Risk	Likelihood	How It Is Handled
System used for triage	Moderate	Acuity and needs are chosen by the user or crew. Clinical judgment overrides the application, every screen carries a direct 911 button, and prominent notices state that the application does not assess patients (Limitation L10).
Fare parameters of Table 6 become outdated	Moderate	Fares sit in one dated configuration file, and every downstream result is recomputed by script from it, with no manual entry.
Mobility class misjudged by the user	Moderate	Only two clearly described classes; the walking time of every leg is shown so the user can judge (Limitation L17).
Surge assignment treated as an order rather than advice	Low	The assignment is shown only to a dispatcher, who confirms each placement. It uses only validated bed reports and is never sent to patients automatically (Limitation L16).
Cost diverts an emergency patient from the nearest capable hospital	Low	Acuity levels 1 and 2 bypass the payment filter (Definition 17; Republic Act No. 10932, 2017). The gap was zero in all 80,000 simulated checks (Table 15).
Site-selection output read as a budget decision	Low	Results are labeled as model outputs; Definition 26 states what is and is not counted, and costs are not modeled (Limitation L18).
Forged or spoofed reply received	Low	Rule V1 checks the sender and a one-time token. All 200 attempted forgeries were rejected.
Personal information received by mistake	Low	Inquiries ask only for aggregate counts. Any reply containing extra details is quarantined for human review, and non-aggregate text is deleted (Republic Act No. 10173, 2012).
Loss of code or data	Low	Version control with remote backups. Every reported figure can be traced to a specific script, seed, and input checksum.

APPENDIX F. THE INQUIRY MESSAGE AND ITS REPLY FORM

The message template of the Inquiry agent is shown below.

Subject: Request for bed status [BSI-7c2e90d4a1b8]
To: the facility’s published official address

Good day.

This message was sent automatically by a bed-availability service built as a student research project at Legazpi City Science High School in Legazpi City; our research adviser is Dick Anthony M. Calleja. We built it so that ambulance crews and patients in Albay can find a hospital with a bed available.

Only three items are needed, and no patient information should be included. Kindly answer by replying to this message. Please keep the code in the subject line as it is and complete the three lines below:

AUTHORIZED BED CAPACITY:
BEDS NOW OCCUPIED:
EMERGENCY ROOM STATUS: ACCEPTING / FULL

The app will show your answer as free beds, together with the time you replied.

You will get this request at most three times daily, and never more than one reminder after each. To receive it only once a day, reply ONCE A DAY; to stop it, reply REMOVE US.

Thank you very much.

[project e-mail address] | Bitano, Legazpi City | Legazpi City Science High School

The Reader accepts the reply form in capital or small letters and also free text in English or Filipino.

APPENDIX G. REPRODUCING THE NEW RESULTS

Tables 16, 19, 28, 29, and 30 and Figure 4 are produced by `verify_patient_extensions.py`, kept in the project repository with its output file `results.json`.

1. recomputes Tables 22 and 27 from the travel times of Table 21 as a consistency check, then computes Tables 28 to 30 for instance S1;
2. generates random road graphs, hospitals, and patient profiles, and checks Propositions 9, 10(iv), 14, and 17 against Floyd-Warshall distances, per-vertex searches, and segment-by-segment closures;
3. generates random anchor-hospital instances and checks Propositions 15, 16, and 18 against direct summation, the assignment solver of scipy together with two simple allocation rules, and exhaustive search over all subsets; and
4. generates random timetables with hospital sinks and checks Propositions 9 and 13 in patient mode.

Every number reported from the script is a model output on synthetic or invented inputs and says nothing about real hospitals in Albay.

ETHICS STATEMENT

The study is computational. No human subjects took part and no personal data were gathered.

Road, transit, landmark, hospital, and population data come either from the public sources of Appendix A or from synthetic instances made for verification and plainly marked, such as instance S1 and the mock mail server of Section 3.3. All sources are used on their published terms, and OpenStreetMap data carry the attribution that the Open Database License requires (OpenStreetMap contributors, 2026).

The protocol and the message in Appendix F must be approved by the Scientific Review Committee before any real hospital receives a message. Messages will go only to addresses that facilities publish themselves.

Data integrity was a priority throughout. None of the figures in this paper is a real clinical or operational measurement.

The first research plan asked volunteers to trace routes in the field with GPS. The fully computational approach of Section 3.1 replaced it, and the change was recorded in the Research Plan Addendum.